ENERGY SAVER NORTH CAROLINA (ESNC)
Rebate Application Technical Guide

Homeowners Managing Efficiency Savings (HOMES)
Home Electrification and Appliance Rebate (HEAR)

Introduction: **1**
Overview of the Quality Install Tool (QIT): **3**
Working with photos 3
Install Check step (QIT) **6**
Creating a Project 7
Adding a Workflow 9
Filling out Tabs 10
Finishing an Install 10
Workflow Specific Guidance 12
Heat Pump Water Heater (HOMES and HEAR) 13
High Efficiency Water Heater (HOMES only) 14
Heat Pump Ducted (HOMES and HEAR) 16
Heat Pump Ductless (HOMES and HEAR) 24
High Efficiency Gas Furnace (HOMES only) 29
High Efficiency Modulating Boiler (HOMES only) 31
Electric Cooking (HOMES and HEAR) 33
Heat Pump Clothes Dryer (HOMES and HEAR) 35
Electric Panel Upgrade (HEAR only) 37
Electric Wiring Upgrade (HEAR Only) 39
Duct Air Sealing and Insulation (HOMES and HEAR) 41
Insulation, Air Sealing, Ventilation (HOMES and HEAR) 44
Attic Air Sealing and Insulation 44

Floor Air Sealing and Insulation above Unconditioned Space..... 48
Foundation Wall Air Sealing and Insulation..... 51
Mechanical Ventilation..... 54
Wall Air Sealing and Insulation (Drill and Fill)..... 56
Full Frame Replacement Windows (HOMES Only)..... 60
Insert Replacement Windows (HOMES Only)..... 62
Closeout step..... 65
HEAR - Electrical Panel (Electric Load Service Center)..... 66
HEAR - Electric Cooking Appliance..... 67
HEAR - Electric Wiring..... 67
HEAR - Heat Pump Clothes Dryer..... 68
HEAR - Heat Pump for Space Heating or Cooling..... 68
HEAR - Heat Pump Water Heater..... 68
HEAR - Insulation, Air Sealing & Ventilation..... 69
Line Items (Materials and Equipment)..... 72
Safety Checklist (fill out here not in QIT)..... 73
Invoice..... 76
Project Summary..... 76
Appendix A: Portal Status Meanings..... 78
Appendix B: Invoice Guidance..... 79

Introduction:

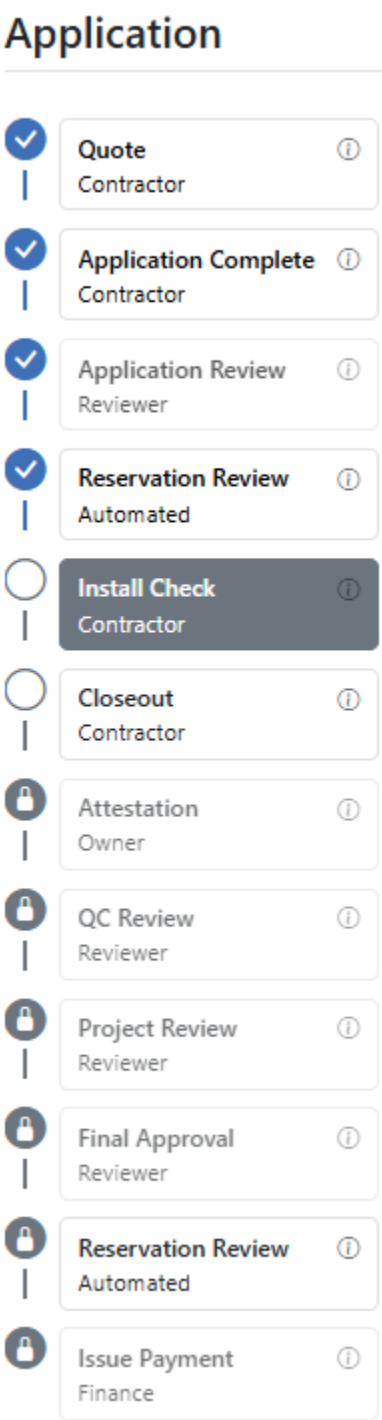
The Energy Saver NC Portal is organized into sections in the order you will enter your application information.

This guide concentrates on the **Install Check Step** and **Closeout** steps. The steps up until these are covered in the Rebate Reservation Guide. The steps after the Closeout step are all steps performed by program staff.

Most of this document is a walk-through of the Pacific Northwest National Laboratory's (PNNL) Quality Install Tool (QIT). This tool is embedded into the Energy Saver NC's portal at the Install Check step. See their website for more information. **While we recommend exploring the tool on their website, for submission purposes you will need to use the tool embedded in the program's portal, we will not accept projects from the PNNL site.**

Acronyms and Abbreviations:
ESNC: Energy Saver NC
DEQ: Department of Environmental Quality
DOE: Department of Energy
HPWH: Heat Pump Water Heater
IDU: Indoor Unit
ODU: Outdoor Unit
PNNL: Pacific Northwest National Laboratory
QIT: Quality Install Tool

Energy Saver NC Portal step overview:
Please note that the steps indicate who is responsible for its completion. Contractor steps are your responsibility. Reviewer, Owner, Automated and Finance are handled by other parties.



Quote is where you can upload your quote (estimate or invoice), either at rebate reservation or prior to the assessment being completed.
Application Complete is the Rebate Reservation process, covered in a separate guide.
Application Review is a program staff step to review the contractor submitted application.
Reservation Review is a program step to review the submitted reservation and reserve your funds with the DOE prior to installation on the job.
Install Check is where you fill out the QIT and upload your geotagged, timestamped photos.

Closeout is where you give us the rest of the job, safety and invoice information.
Attestation is a step completed by the customer attesting to the fact that the work was completed.

QC Review is the step in which program staff conduct an in field inspection and/or file review of the job.

Project Review is the step in which the program staff reviews the post installation submission.

Final Review is a secondary review by program staff.

Reservation Review is a program step to review the submission and release your funds for payment from the DOE.
Issue Payment payment is issued through our payment processor.

Overview of the Quality Install Tool (QIT):

The Quality Install (QI) tool is our primary tool to capture information about the work performed and document the before and after. This tool was created as a convenience for all states to use, but is not tailored to this program, so not all data points are required for Energy Saver NC. This guide outlines all required inputs and indicates whether each item is mandatory, optional, or conditionally applicable. Please refer to the specific instructions in this guide for each measure. By providing the required information and following the guidance for testing and photos, you will support timely closeout and payment of the project.

The QI tool is made up of four distinct sections:

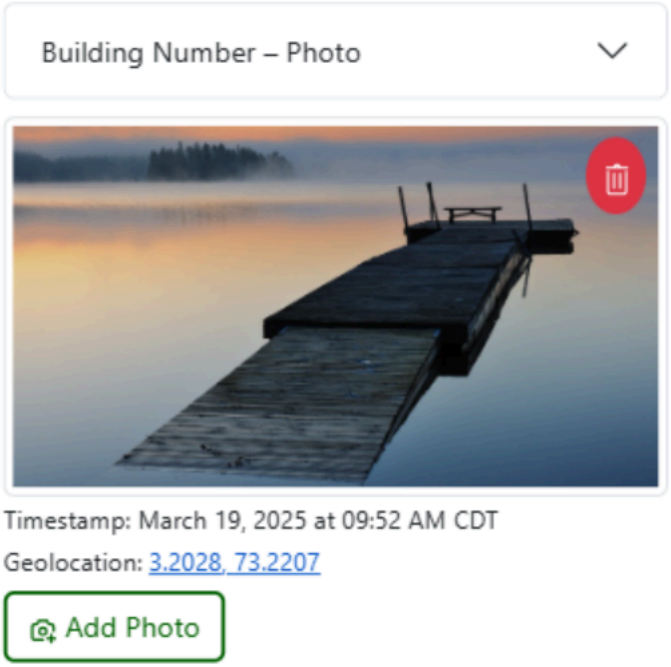
1. **Projects** - The initial (home) screen allows the user to create new projects, and rename, delete, or open existing projects. Adding a new project results in a page to capture a project's name, installer information, and project address.
2. **Workflows** - Within a project, the user can select from the list of workflows that indicate an installation type or situation to add to the project profile.
3. **Installations** – Within a workflow, you add installations. This setup allows you to add multiple installations if you're installing multiple systems of the same type, such as running electrical wire to multiple appliances or adding multiple ductless heat pumps.
4. **Tabs** – Within a workflow, there may be different tabs listed across the top, including Assessment, Planning, Installation, Post-Installation, and Report. It's important to understand what's required on each of these tabs BEFORE starting work so that when required, pre-existing equipment is documented before being removed. See each measure for what tabs populate for each workflow.

Working with photos

Every project and almost all workflows require a number of photos. Photos submitted should all be geotagged, which reflects that the picture was taken at the

matching project address. Geotagging requires a device with GPS or a cellular connection that can identify the coordinates of the device and the device's settings must be configured to tag the photo with those coordinates, almost any smartphone and most tablets have this capability, as do many standalone cameras. You should verify that your devices are configured to geotag before beginning your first project. Here are some resources for working with geotagged photos:

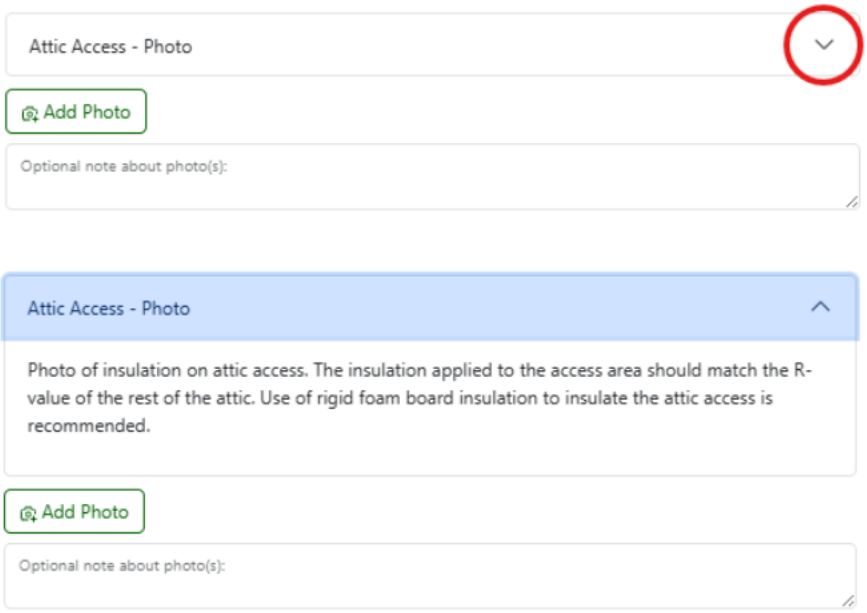
- Capturing Geotagged Photos
- Transferring Geotagged Photos
- Viewing Geotagged Data in Photos



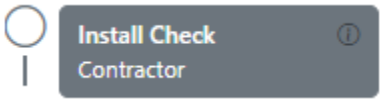
A properly geotagged photo will display GPS coordinates once it is uploaded. The Quality Install Tool pulls the coordinates from the device being used to upload the photo. This means that if photos are uploaded from an office computer, the system may show the computer's location instead of the job site.

As long as the photo itself contains the correct embedded coordinates from when it was taken, this is acceptable. You can click on the displayed geolocation coordinates to open a map and verify the location.

Note that everywhere you can add a photo, you can add multiple photos sequentially. When a photo is required, there is usually a drop-down description that can provide more detail about what should be in the photo like this:



Install Check step (QIT)



This tab of the Energy Saver NC's portal is where the QIT is embedded. This section will take you through the data entry process for each measure. Keep in mind that all data inputs that you will see while in the QIT are listed here, but **not all available entries in the QIT are required for rebate submission to Energy Saver NC.**

The entries in this guide that are marked with Not Required do not have to be filled in for rebate submission for this program.

PHOTO: is a prompt to upload a photo, followed by the description you will see in the tool and sometimes some clarification of that description.

Many entries are only required if applicable to your project. An example of this is air sealing locations; not every air sealing job will have a photo of knee walls or skylights. If that prompt is not applicable to your project it is not required and may be skipped. When in doubt, err on the side of submitting a photo, more information is always appreciated and helpful. **Skipping necessary information will result in processing/payment delays.**

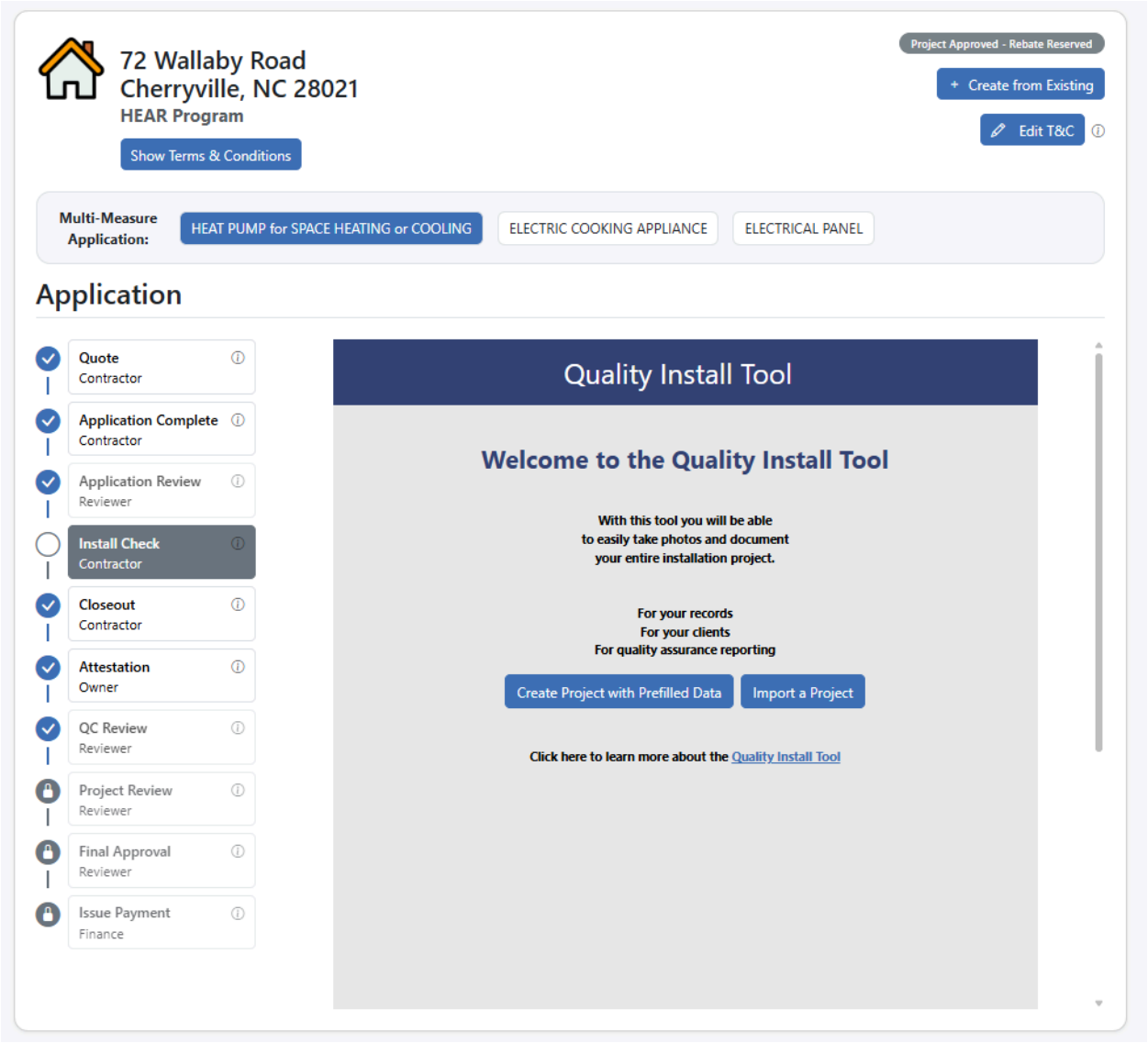
CFM at 50Pa entries, or blower door test results, are not required for this program but may be used. Our assessors do not perform performance testing within this program. If there is an actual blower door result it can be input and accompanied by the corresponding geotagged, time-stamped photos. Otherwise, choose "Not measured".

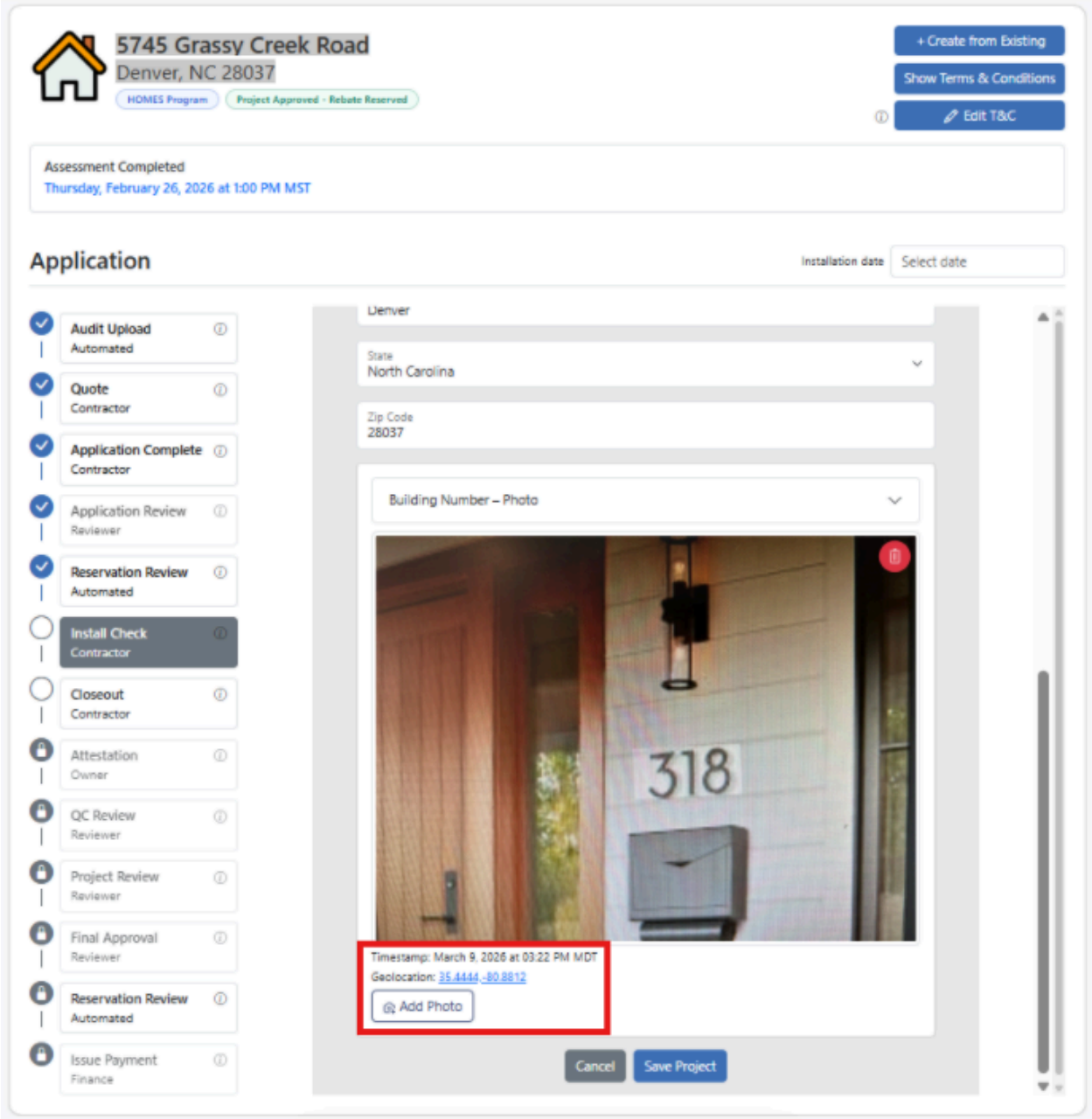
Combustion Safety Testing is reported in our portal in the closeout tab, not within the QIT. Combustion safety testing is required if there is natural draft heating or water heating equipment left post-improvement.

HOMES vs HEAR rebate entry in the QIT. The QIT does not identify which program the measure is for, so regardless of which rebate path you take the entries for that measure are the same. For example, the QIT entries for a Heat Pump water heater (HP WH) will be the same regardless of if you are applying for a HOMES or HEAR rebate.

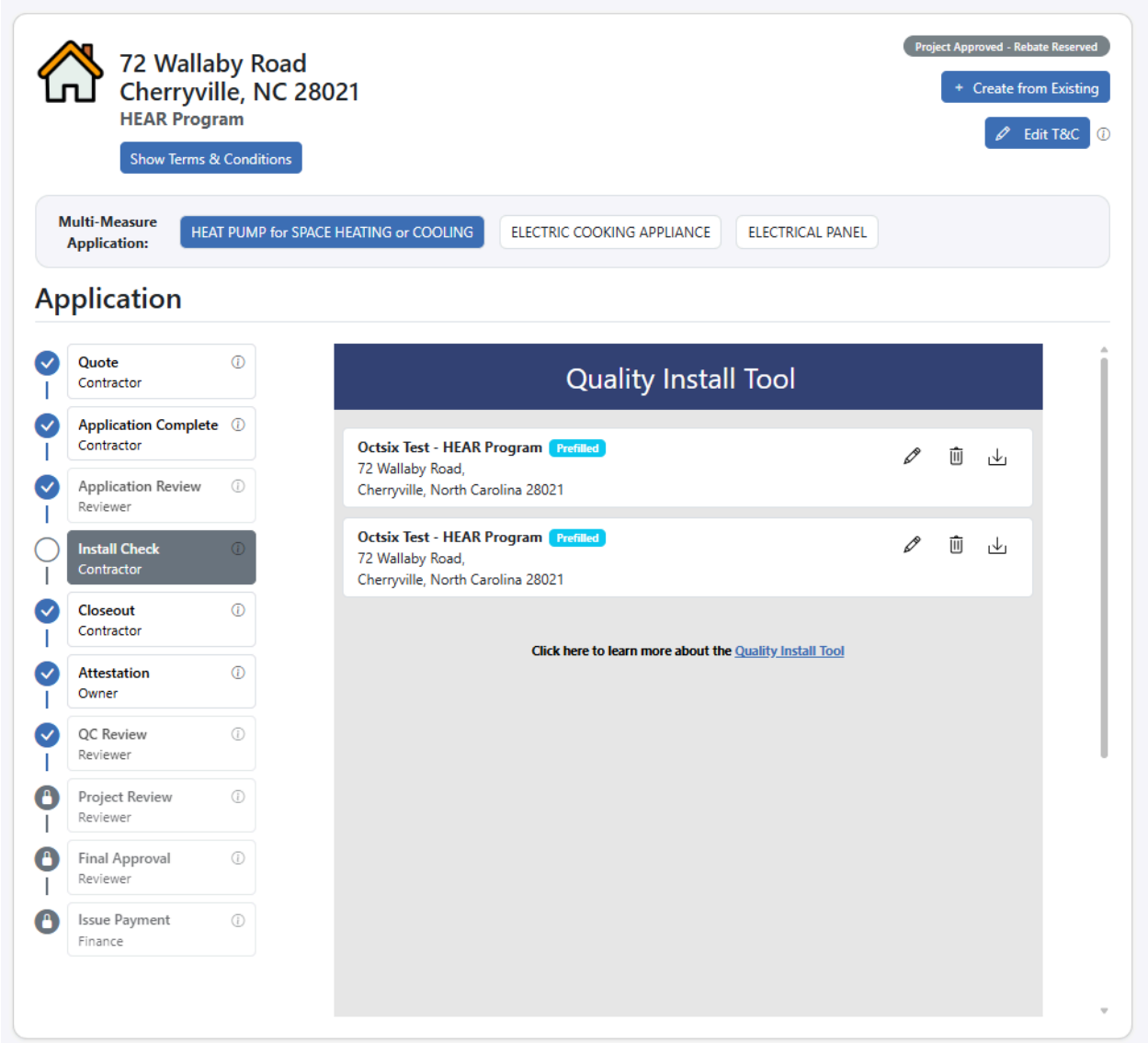
Creating a Project

To begin the post installation process click on Create a Project with Prefilled Data





- Scroll through and make sure all information is correct; edit it if necessary
- PHOTO: Building number
    - After uploading the first photo, verify that the geotag and timestamp are displayed beneath it. Clicking on the geotag location will confirm the correct location is associated with the photo. Note the latitude and longitude numbers to confirm this for every photo upload, as this is how you ensure the feature is functioning properly.
- click Save Project



Click the pencil to edit Project Info that you just reviewed. Click on Name/Address to enter the QIT and begin adding workflows.

Adding a Workflow

Next you'll choose the installation workflows to add to the project. To add a workflow, just click on it. This will present you with the Installations screen. Click on Add Installation.

What to do if there is more than one system?

Each workflow can be used more than once; for example, a home might have two separate ductless heat pumps (each serving a different area) or more than one attic to air seal and insulate. This is the purpose of having Installations under each

workflow. Simply add an installation for each separate system you're installing. You may need to enter some of the same data for each installation – please do not leave a field blank on one installation just because it was already entered for another.

Filling out Tabs

The tabs are organized left-to-right to match the flow of the project from gathering information to completing the work and verifying the equipment runs properly. Each tab has different sections and are laid out so that you can start at the top and work your way down.

Wherever there is a title with drop down arrow at the far right of a line, you can click on it to expand the details about that section. This provides some on-the-job guidance to help with collecting the right information and avoiding safety issues or failed QC inspections. For example, on the Attic Air Sealing and Insulation workflow on the Pre-Installation tab, you can expand the Existing Conditions section which notes that it is not acceptable to just add more air-permeable insulation to an attic without first air-sealing.

On lines that have a question with drop down arrow at the right, clicking on the line will present you with choices to answer the question. You must choose an answer, whether a drop down or radio button, for every line unless it says that it is optional or we've noted that it's not needed in this guide.

When you reach the bottom of the tab, go back to the top and click on the next tab. You'll likely have to come back to certain tabs as you go through the installation process. Any time you enter data it is automatically saved.

What happens when I click the back button?

At any point, if you click the back button in the top left corner, it will take you "up a level", meaning it will take you to the installations when you're on a tab, to the workflows when you're on installations, and to the project when you're on workflows.

Finishing an Install

When you are done filling out all of the workflows, you can simply close the QI tool. All of the data is saved to the application and will be provided to the rebate processing team and Quality Control (QC) analyst when you finish attaching documents and other required data and submit the project for closeout. The QC analyst or rebate processing team may reach out to you with questions about the

QI documentation if something is missing or looks like it may be wrong. You can refer to the Contractor Manual for guidance on how issues might be handled.

For all installations that have a “photo of invoice” upload section, you do not need to complete that in the QI tool. There will be a separate upload of the final paid invoice requested during the project closeout submission within the portal, not the QI tool.

Remember to leave all relevant warranty information and product manuals with the customer or show them where they are located if contained in a pouch attached to the installed equipment. Also, thank the customer for their participation in the North Carolina Energy Savers Program and choosing you!

Workflow Specific Guidance

For each workflow, we'll go through the tabs and provide guidance on completing the tabs. You can find additional guidance within the tool. If you ever have a question while you're in the field, please reach out to the program's contractor manager for guidance. They may answer on the spot or need to consult with a program specialist, but they will always help get the answer you need.

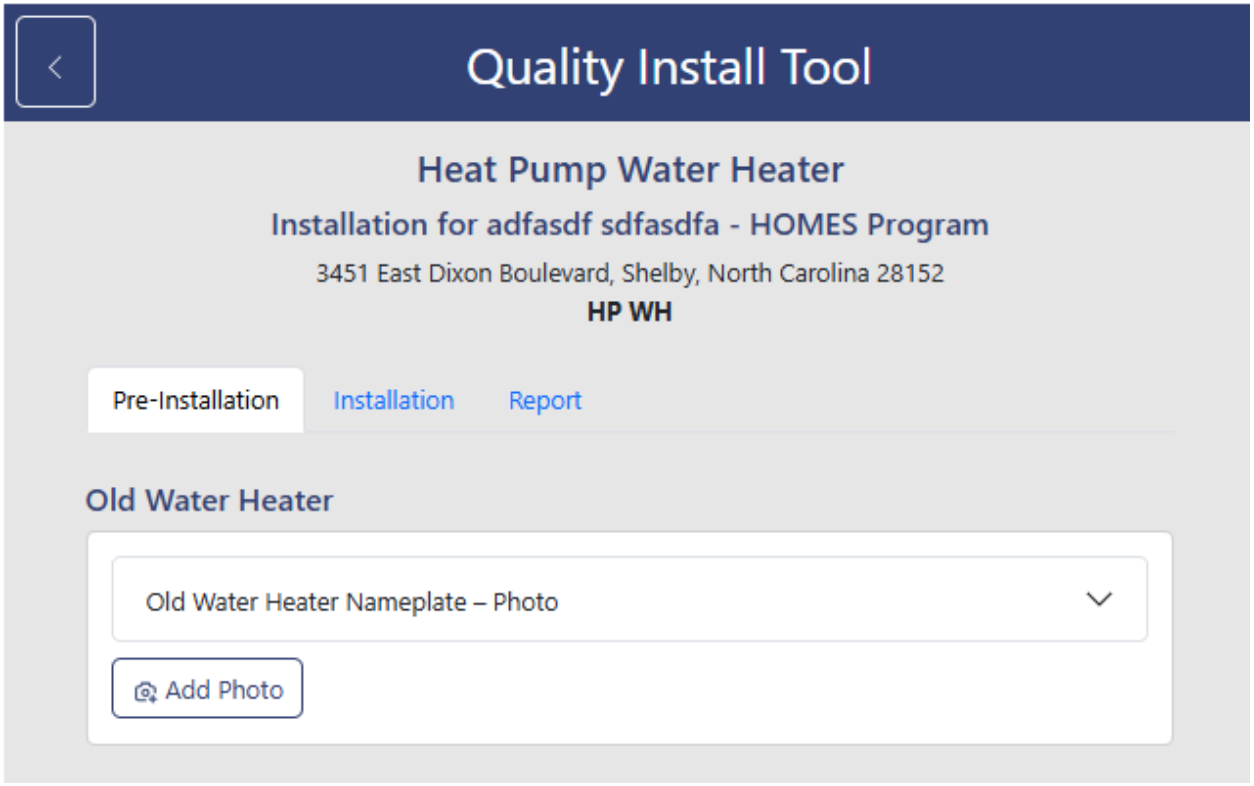
Choose an Installation Workflow from:

- Heat Pump Water Heater (HOMES and HEAR)
- High Efficiency Water Heater (HOMES only)
- Heat Pump Ducted (HOMES and HEAR)
- Heat Pump Ductless (HOMES and HEAR)
- High Efficiency Gas Furnace (HOMES only)
- High Efficiency Modulating Boiler (HOMES only)
- Electric Cooking (HOMES and HEAR)
- Heat Pump Clothes Dryer (HOMES and HEAR)
- Electric Panel Upgrade (HEAR only)
- Electric Wiring Upgrade (HEAR Only)
- Duct Air Sealing and Insulation (HOMES and HEAR)
- Insulation, Air Sealing, Ventilation (HOMES and HEAR)
    - Attic Air Sealing and Insulation
    - Floor Air Sealing and Insulation above Unconditioned Space
    - Foundation Wall Air Sealing and Insulation
    - Mechanical Ventilation
    - Wall Air Sealing and Insulation (Drill and Fill)

Heat Pump Water Heater (HOMES and HEAR)

Add installation

Enter New installation name



Pre-Installation Tab

Please pay attention to the space evaluation and building science sections, especially if you're newer to installing heat pump water heaters, as there are more considerations than just capacity and access to sufficient air.

- Old Water Heater
    - PHOTO: nameplate of existing equipment
- HPWH Space Evaluation:
    - Volume of space: (enter volume of space) Not required
    - PHOTO: HPWH Location (photo of space to prove space requirements) Not required
    - PHOTO: Adjacent space (if used for space requirements) Not required

Installation Tab

- Installation Date Not required

- PHOTO: Completed installation (overall of new equipment showing exhaust, insulated water lines, pressure relief valve discharge and condensate line)
- PHOTO: nameplate and EnergyGuide sticker of new equipment
- PHOTO: HP WH screen with the mode Not required
- PHOTO: CTA-2045 module, if installed Not required
- Combustion Appliance Safety Testing (Do not report here, reported in Closeout step)
    - If the home upgrades will result in a barometric draft controlled or Category I appliance equipped with a draft hood or connected to a natural draft venting system remaining in the conditioned envelope of the home, combustion safety checks are required to be performed at the completion of the project.

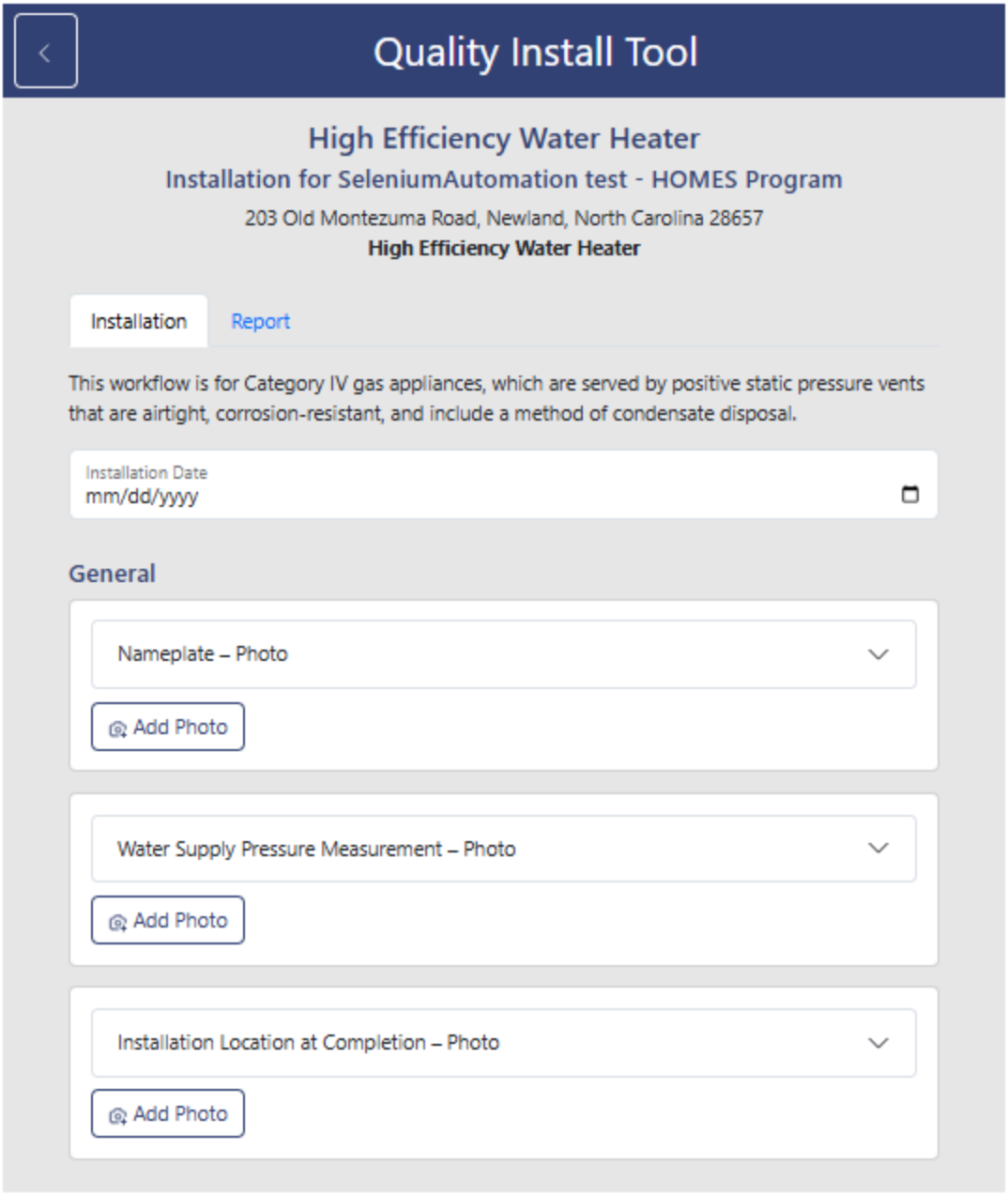
Report tab

This tab allows you to see all inputs from other tabs. If all information is correct and complete, click Submit Final Report. Once you submit your report, your entries cannot be accessed or changed. **If your application is returned, you will have to resubmit all information within the QIT.**

High Efficiency Water Heater (HOMES only)

Add installation

Enter New installation name



Installation tab

- Installation Date Not required
- General
    - PHOTO: Nameplate and EnergyGuide sticker of new equipment
    - PHOTO: Water supply pressure measurement Not Required
    - PHOTO: Completed installation photo (overall of new equipment showing exhaust, insulated water lines, pressure relief valve discharge and condensate line)

- Venting
    - PHOTO: Exhaust termination at exterior showing clearance from windows, doors, air intakes, crawlspace vents, etc. Not Required
    - PHOTO: Intake, if not shown in other photos Not Required
    - PHOTO: Remaining natural draft appliances, if applicable Not Required
- Condensate
    - PHOTO: Condensate management photo if not in overall Not Required
- Gas
    - PHOTO: Carbon monoxide detector installed Not Required
    - PHOTO: Electronic Methane Leak Detector Not required
    - PHOTO: Combustion analysis results Not required

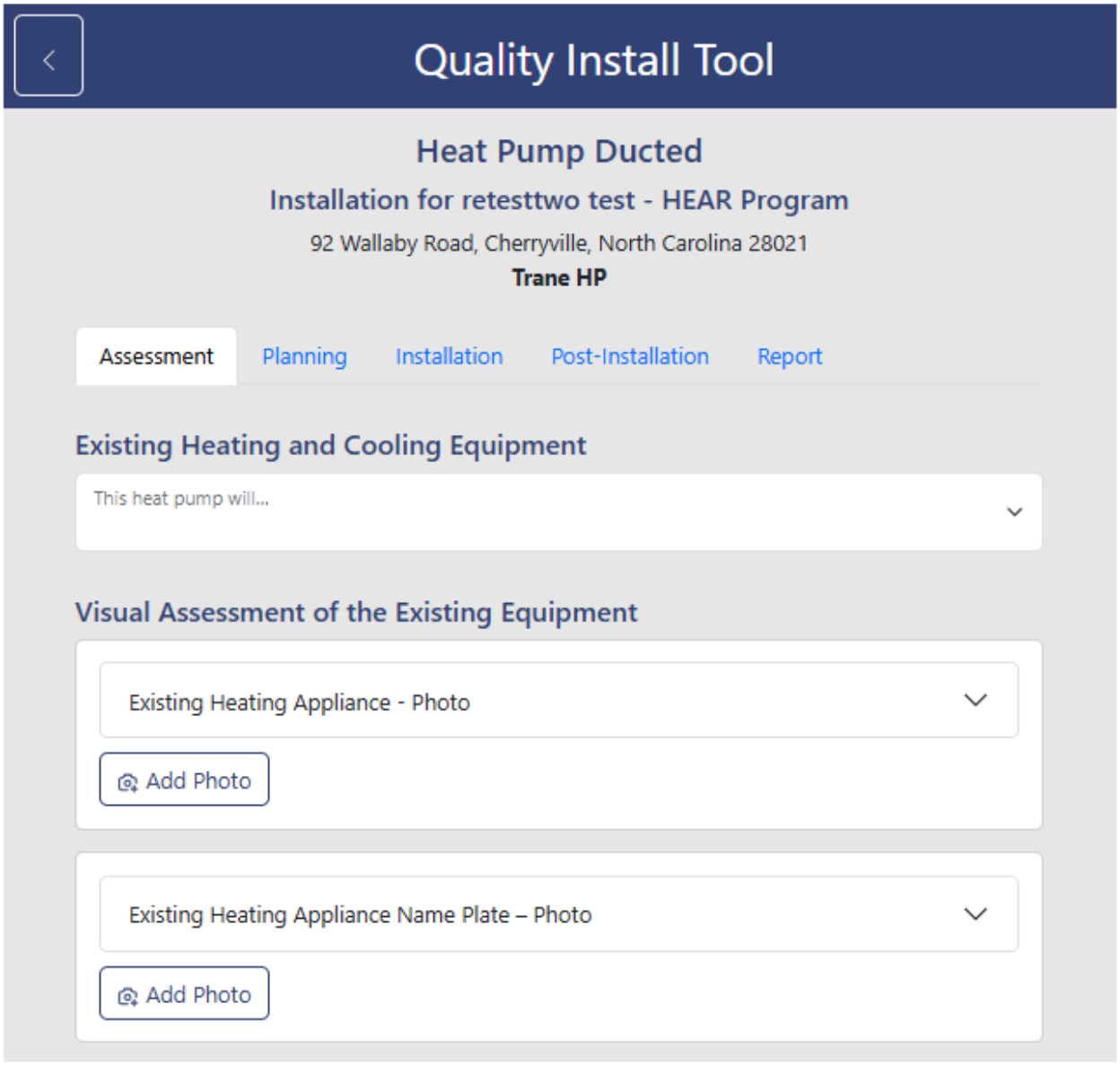
Report tab

This tab allows you to see all inputs from other tabs. If all information is correct and complete, click Submit Final Report. Once you submit your report, your entries cannot be accessed or changed. **If your application is returned, you will have to resubmit all information within the QIT.**

Heat Pump Ducted (HOMES and HEAR)

Add installation

Enter New installation name



Assessment tab

Airflow Test – Not required, but if you can access the airflow data for the equipment, please use your total static pressure results and the airflow data to identify the CFM. A better understanding of the current ductwork's effect on performance can help you address performance issues that aren't fixed just by installing the new system.

If not testing the airflow pre-upgrade, please at least note how you plan to achieve airflow of 350-450 cfm per ton post-upgrade while keeping static pressure below the maximum operating range of the equipment.

The entire point of the assessment section is to evaluate the ductwork and if any duct repairs or upgrades are needed so that the heat pump yields the expected efficiency and comfort benefits.

- Existing Heating and Cooling Equipment: This heat pump will…
    - Replace electric resistance heat
    - Replace natural gas heat
    - Replace fuel oil heat
    - Replace propane heat
    - Replace wood heat
    - Displace/Reduce electric resistance heat
    - Displace/Reduce natural gas heat
    - Displace/Reduce fuel oil heat
    - Displace/Reduce propane heat
    - Displace/Reduce wood heat
- Visual Assessment of the Existing Equipment
    - PHOTO: Overall of existing heating equipment
    - PHOTO: Existing heating equipment nameplate
    - Checkbox: No cooling appliance If cooling equipment exists, check then uncheck this box to populate cooling equipment fields below
    - PHOTO: Overall of existing cooling equipment
    - PHOTO: Existing cooling equipment nameplate
- Visual Assessment of Ductwork (overall of duct system showing connections to plenum, supply and return connections, takeoffs, sizing of ductwork, trunks, boots to surface)
    - PHOTO: Ductwork photo(s)
    - Comment box
- Static Pressure Test
    - PHOTO: Pre-Upgrade Static Pressure Test Not Required
    - Comment box
- Airflow Test
    - PHOTO: Pre-Upgrade Airflow Test Setup Not required
    - PHOTO: Pre-Upgrade Airflow Test Results Not required
    - Comment box
- Duct Leakage Testing Not required

    - Select the test method used: Duct Leakage tester, Pressure Pan, Airflow Hood Not required
- Combustion Appliance Safety Testing (Do not report here, reported in Closeout step)
    - If the home upgrades will result in a barometric draft controlled or Category I appliance equipped with a draft hood or connected to a natural draft venting system remaining in the conditioned envelope of the home, combustion safety checks are required to be performed at the completion of the project.

Planning tab

Compressor lockout on an all-electric system should be set to manufacturer's requirements to avoid damage to the compressor or when system COP is approaching 1.0. Often, you will never see temperatures that reach the compressor lockout setting. On a dual-fuel system, compressor lockout should be set based on the customer's desire. If they desire to minimize cost, the compressor lockout should be set to the economic balance point – the same as the switchover temperature. If they wish to maximize the use of their heat pump, set the lockout as noted previously for an all-electric system.

Aux heat lockout on an all-electric system should be set to no more than 5° F above the comfort balance point. This should generally be no higher than 35° F.

For a dual-fuel system, the *switchover temperature* must be set such that at a minimum, the heat pump provides the majority of heating energy for the home. Beyond that, you can set the switchover temperature based on the homeowner's desire. If they want to minimize cost above all else, then set the switchover temperature to the break-even COP based on the electricity rate and fuel cost (i.e. economic balance point) or to the minimum setting so the heat pump is still providing the majority of heating. If they would like to maximize use of their heat pump system (for instance if they have solar or to minimize carbon emissions), then the switchover temperature should be set to the balance point (i.e. comfort balance point).

Depending on the thermostat and/or heat pump, these settings may be controlled differently. In all cases, it's important to remind your customers not to use setbacks with their heat pump. Some thermostats will turn on the backup heat if the delta

between the setpoint and ambient temperature is more than 4° F (or a user-configurable setting). This can cause unexpected cost-increases and decreased comfort.

- Planned Install Location
    - PHOTO: Proposed ODU location Not Required
    - ODU Mounting Style Not Required
        - Pad
        - Ground Stand
        - Wall Mount
        - Roof Mount
    - ODU inches above the ground (elevated above the snow): Not Required
    - Overhead Snow & Ice Protection Not Required
        - Awning/Cover
        - None
- Manual J Calculation Not required (uploaded in closeout step)
    - Upload Manual J (pdf)
    - Comment box
- Equipment Performance Tables (please input lockout and switchover temps)
    - PHOTO: Mfr's Heating Performance Table Not required
    - Aux Heat Lockout Above this Temperature:
    - Compressor Lockout Below this Temperature:
    - Dual Fuel Switch Over Temperature:
    - PHOTO: Mfr's Cooling Performance Table Not required
- Pre-installation Additional Comments
    - Comment box

Installation tab

- Installation Date Not required
- Refrigerant Recovery
    - Is this heat pump installation replacing an old AC or heat pump? Yes/No
- Equipment
    - PHOTO: New ODU nameplate and EnergyGuide sticker
    - PHOTO: New IDU nameplate and EnergyGuide sticker

- Electrical
    - PHOTO: ODU circuit breaker Not required
    - PHOTO: IDU circuit breaker Not required
    - Circuit Breaker notes box
- Installation Tests
    - PHOTO: Upload Nitrogen Pressure Test Setup Not required
    - PHOTO: Upload Nitrogen Pressure Decay Test Results (Claims may be denied if pressure shown is not at least 250 or if the photo indicates that pressure dropped more than 5 psi over a half hour when adjusted for temperature. Photo should show the time, pressure and temperature deltas). Example photo:



    - PHOTO: Upload Vacuum Decay Test Setup Not required
    - PHOTO: Upload Vacuum Decay Test Result (500 microns or less held for at least 10 minutes. Digital gauge which displays the time, pressure, and delta in pressure or a gauge showing the pressure under 500 and a timer in the photo is acceptable) Example photo:



- Refrigerant Adjustments
    - PHOTO: Upload Refrigerant Adjustments photo, if applicable Not required
    - Additional Refrigerant Calculator Not required
        - Feet of line set beyond factory charge:
        - Ounce of refrigerant per foot of line set:
    - Notes box
- Protection
    - PHOTO: Condensate Line Not required
    - PHOTO: Line Set and line set wall penetration (insulation protected from weather and UV)
    - PHOTO: Electrical Surge protection device, if applicable Not required
    - PHOTO: Service Caps Reinstalled and Tight, if applicable Not required
- Plenum Connections
    - PHOTO: Plenum Connections photo (supply and return connections to air handler, sealed and, if applicable insulated)

Post-installation tab

You should have the airflow data for the new system and be able to complete the airflow test using the static pressure measurements even without a flow plate. This is necessary to verify that the new system is installed to operate within manufacturer specifications to deliver the rated capacity.

For dual-fuel systems:
Dual-fuel mode should be enabled to ensure the heat pump and backup furnace or boiler do not run at the same time.

The switchover temperature is set using the **compressor lockout** setting (sometimes referred to as low balance point or minimum outdoor temperature). It needs to be set such that, at minimum, the heat pump is covering more than 50% of the home's heating load, per DOE's program requirements. It should also be set no higher than the economic balance point (i.e. break-even COP) and no lower than the thermal balance point where the heat pump's capacity equals the home's heat load. If the setting is above 35, please provide proof of calculations that demonstrate the requirements are met.

For all-electric systems with auxiliary strip heat:
The **compressor lockout** should either be disabled or set to the heat pump's minimum operating temperature defined by the manufacturer.
The **auxiliary lockout** should be set a few degrees above the thermal balance point where the heat pump's capacity equals the home's heat load. This can be set as high as 5 degrees above the thermal balance point for comfort and recovery. This setting should never be above 35, but ideally much lower. The goal is to minimize the use of auxiliary heat and therefore minimize negative impact to residents' energy bills.

Reminder that the thermostat should not be programmed with setback temperatures and the occupants should be educated on how temperature setbacks at night or when away for periods less than a few days can negatively affect their energy usage.

- Post-install Airflow Test
    - PHOTO: Post-install Airflow Test Setup Not required
    - PHOTO: Post-install Airflow Test Results Not required
- Post-install Static Pressure Test: If it is out of bounds this could delay rebate payment
    - PHOTO: Post-install Static Pressure Test Setup Not Required
    - PHOTO: Post-install Static Pressure Test Results (supply and return photos required. 0.55” WC or less or provide manufacturer’s documentation that result is allowable)
- Thermostat Settings

    - PHOTO: Upload Thermostat Setting (showing aux heat lockout, compressor lockout, duel fuel switchover temps. For systems with strip heat, the auxiliary lockout is 35 or lower. For dual-fuel systems, compressor lockout is 35 or lower. If above 35, then proof of calculations provided showing heat pump provides more than 50% of clearly labeled annual heating load and is not higher than the clearly labeled economic balance point.)
    - Comment box
- Project Invoice Not required (uploaded on Closeout tab)
    - PHOTO: Project Invoice
- Mechanical Ventilation
    - If the addition of supply based mechanical ventilation is included in the heat pump upgrade. Use the mechanical ventilation quality installation workflow to document that installation.
- Combustion Appliance Safety Testing (Do not report here, reported in Closeout step)
    - If the home upgrades will result in a barometric draft controlled or Category I appliance equipped with a draft hood or connected to a natural draft venting system remaining in the conditioned envelope of the home, combustion safety checks are required to be performed at the completion of the project.

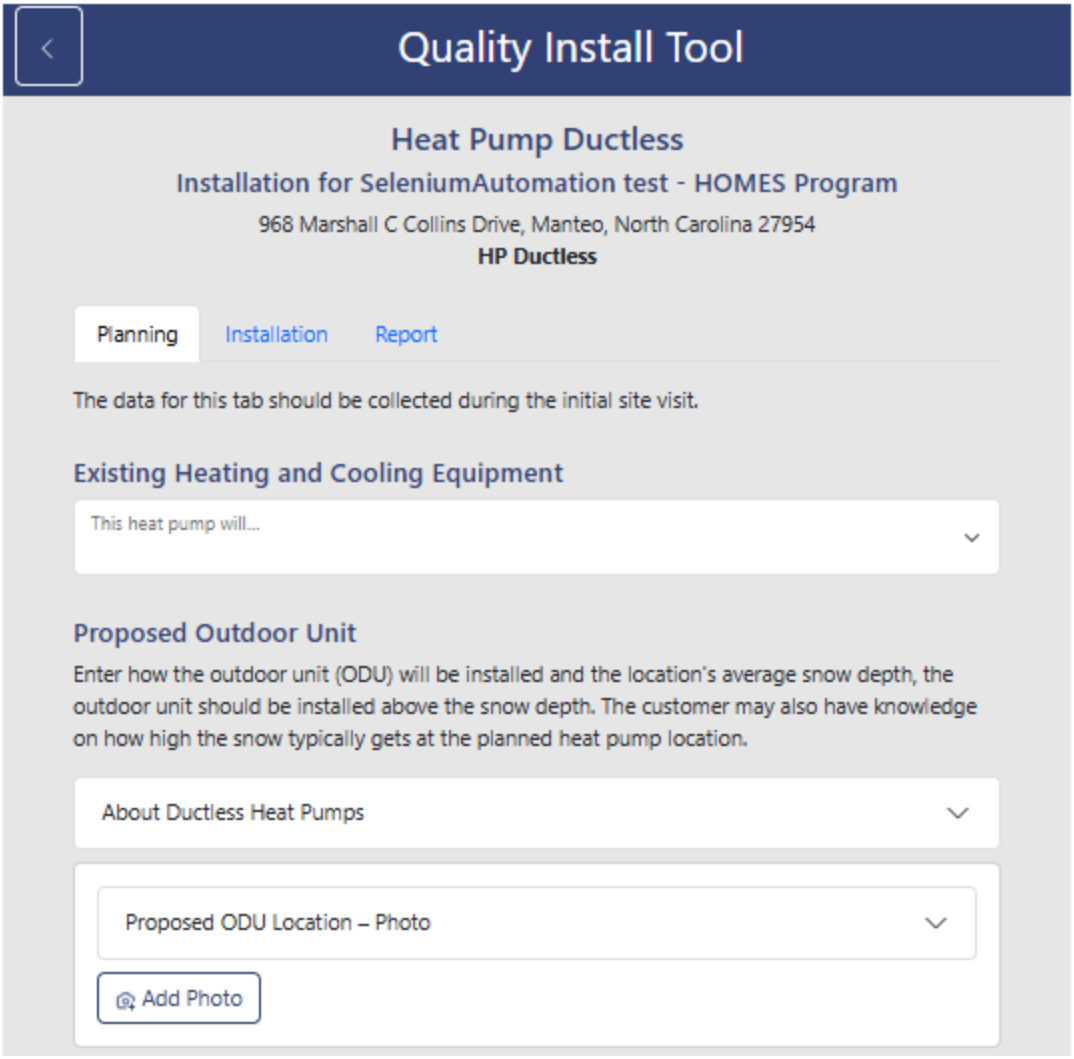
Report tab

This tab allows you to see all inputs from other tabs. If all information is correct and complete, click Submit Final Report. Once you submit your report, your entries cannot be accessed or changed. **If your application is returned, you will have to resubmit all information within the QIT.**

Heat Pump Ductless (HOMES and HEAR)

Add installation

Enter New installation name



Planning tab

- Existing Heating and Cooling Equipment: This heat pump will...
    - Replace electric resistance heat
    - Replace natural gas heat
    - Replace fuel oil heat
    - Replace propane heat
    - Replace wood heat
    - Displace/Reduce electric resistance heat
    - Displace/Reduce natural gas heat
    - Displace/Reduce fuel oil heat
    - Displace/Reduce propane heat
    - Displace/Reduce wood heat
- Proposed Outdoor Unit

    - PHOTO: Proposed ODU Location Not required
    - ODU will be installed on a
        - Pad
        - Ground Stand
        - Wall Mount
        - Roof Mount
    - Average snow depth Not required
    - ODU installation location comments:
- Proposed Outdoor Unit
    - PHOTO: Proposed IDU Location Not required
    - IDU installation location comments:
    - PHOTO: Floor Plan sketch Not required
- Manual J Calculation Not required (uploaded in closeout step)
    - Upload Manual J (pdf)
- Equipment Selection
    - PHOTO: Heating Performance Table Not required
    - PHOTO: Cooling Performance Table Not required
- Combustion Appliance Safety Testing (Do not report here, reported in Closeout step)
    - If the home upgrades will result in a barometric draft controlled or Category I appliance equipped with a draft hood or connected to a natural draft venting system remaining in the conditioned envelope of the home, combustion safety checks are required to be performed at the completion of the project.

Installation tab

- Installation Date Not required
- Refrigerant Recovery
    - Is this heat pump installation replacing an old AC or heat pump? Yes/No
    - PHOTO: new ODU nameplate and EnergyGuide sticker
    - PHOTO: new IDU nameplate and EnergyGuide sticker
    - PHOTO: Circuit breaker amperage Not required
    - PHOTO: Flare tested in a flare gauge Not required
    - PHOTO: Torque wrench setting Not required

- PHOTO: Upload Nitrogen Pressure Test Setup Not required
- PHOTO: Upload Nitrogen Pressure Decay Test Results (Claims may be denied if pressure shown is not at least 250 or if the photo indicates that pressure dropped more than 5 psi over a half hour when adjusted for temperature. Photo should show the time, pressure and temperature deltas). Example photo:



- PHOTO: Upload Vacuum Decay Test Setup Not required
- PHOTO: Upload Vacuum Decay Test Result (500 microns or less held for at least 10 minutes. Digital gauge which displays the time, pressure, and delta in pressure or a gauge showing the pressure under 500 and a timer in the photo is acceptable) Example photo:



    - PHOTO: Installation Manual's line set compensation instructions Not required
    - Ounce of refrigerant to add on top of factory charge.
        - Feet of line set beyond factory charge Not required
        - Ounce of refrigerant per foot of line set Not required
        - PHOTO: Upload Refrigerant Adjustments photo, if applicable Not required
        - Additional Refrigerant Calculator Not required
        - Notes box
    - PHOTO: Condensate Line Not required
    - PHOTO: Line Set Protection, and line set wall penetration (insulation protected from weather and UV)
    - PHOTO: Electrical Surge protection device, if applicable Not required
    - PHOTO: Service caps reinstalled and tight, if applicable Not required
- Project invoice
    - PHOTO: Project invoice Not required (uploaded on Closeout tab)

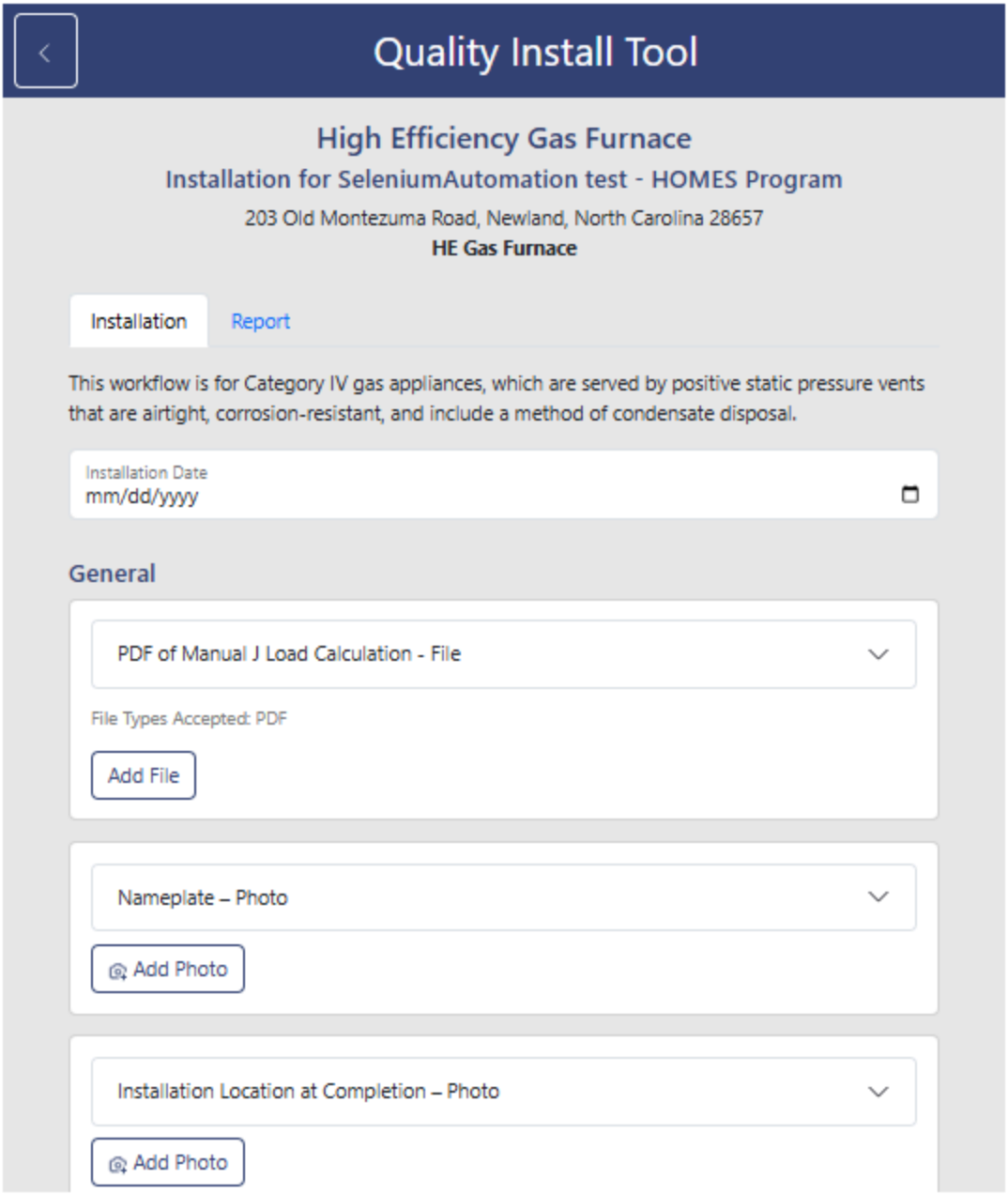
Report tab

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High Efficiency Gas Furnace (HOMES only)

Complete all sections as shown. You should have the airflow data for the new system and be able to complete the airflow test using the static pressure measurements even without a flow plate. This is necessary to verify that the new system is installed to operate within manufacturer specifications to deliver the rated capacity.

Add installation
Enter New installation name



Installation tab

- Installation Date Not required
- General
    - Upload Manual J (pdf) Not required (uploaded in closeout step)
    - PHOTO: New nameplate and EnergyGuide sticker
    - PHOTO: New equipment overall
- Venting (show clearance from openings, eaves, vents, and intakes, correct flashing)

    - PHOTO: Exhaust vent termination Not required
    - PHOTO: Intake vent location (if not in prior photo) Not required
    - PHOTO: Remaining natural draft combustion appliances Not required
- Condensate (connections to combustion air inlet moisture trap, AC coil, acid neutralizer, secondary drain pan and condensate pump if applicable)
    - PHOTO: Condensate management piping Not required
- Airflow
    - PHOTO: Manometer showing external static pressure Not required
    - PHOTO: Airflow measurement Not required
    - PHOTO: Air filter Not required
- Gas
    - PHOTO: Carbon monoxide alarm installed Not required
    - PHOTO: Electronic methane leak detector Not required
    - PHOTO: Manometer showing outlet gas pressure Not required
    - PHOTO: Combustion analysis results Not required

Report tab

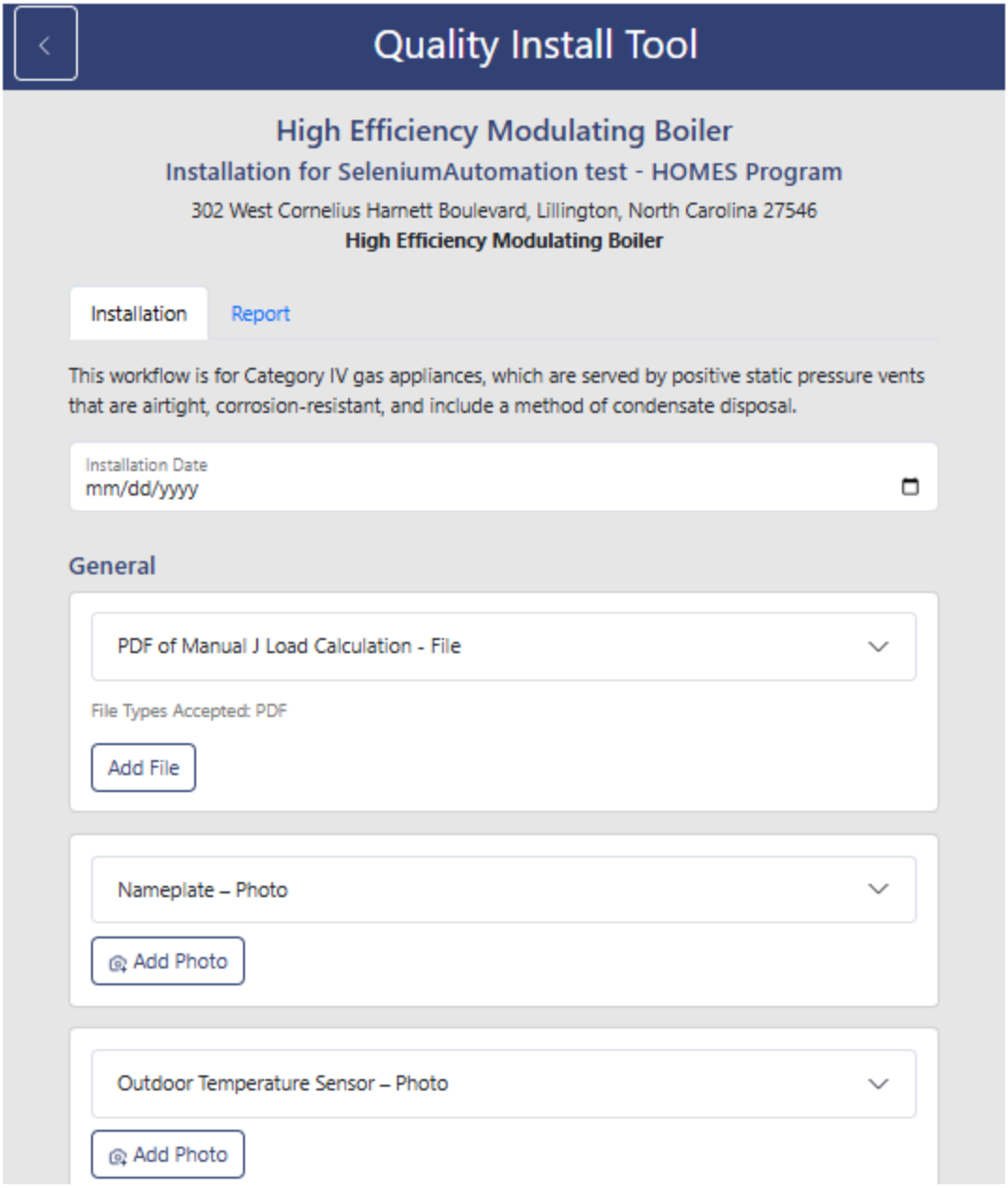
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High Efficiency Modulating Boiler (HOMES only)

(listed as Variable Frequency Drives in Planned Upgrades in Rebate Reservation process)

Add installation

Enter New installation name



Installation tab

- Installation Date Not required
- General
    - Upload Manual J (pdf) Not required (uploaded in closeout step)
    - PHOTO: New nameplate and EnergyGuide sticker
    - PHOTO: Outdoor temperature sensor (located on north wall)
    - PHOTO: New equipment overall
    - PHOTO: System layout diagram Not required

- Venting (show clearance from openings, eaves, vents, and intakes, correct flashing)
    - PHOTO: Exhaust vent termination Not required
    - PHOTO: Intake vent location Not required
    - PHOTO: Remaining natural draft combustion appliances Not required
- Gas
    - PHOTO: Carbon monoxide alarm installed Not required
    - PHOTO: Electronic methane leak detector Not required
    - PHOTO: Combustion analysis results Not required
- Condensate (connections to combustion air inlet moisture trap, AC coil, acid neutralizer, secondary drain pan and condensate pump if applicable)
    - PHOTO: Condensate management piping Not required
- Hydronics and Controls
    - PHOTO: Boiler system fluid chemistry test results
    - PHOTO: Heating curve selected setting for the coldest outdoor temperature day
    - PHOTO: Heating curve selected setting for the warmest outdoor temperature day
    - Hydronic loop's pressure loss (feet of head)
    - Setting percent maximum pump speed Not required

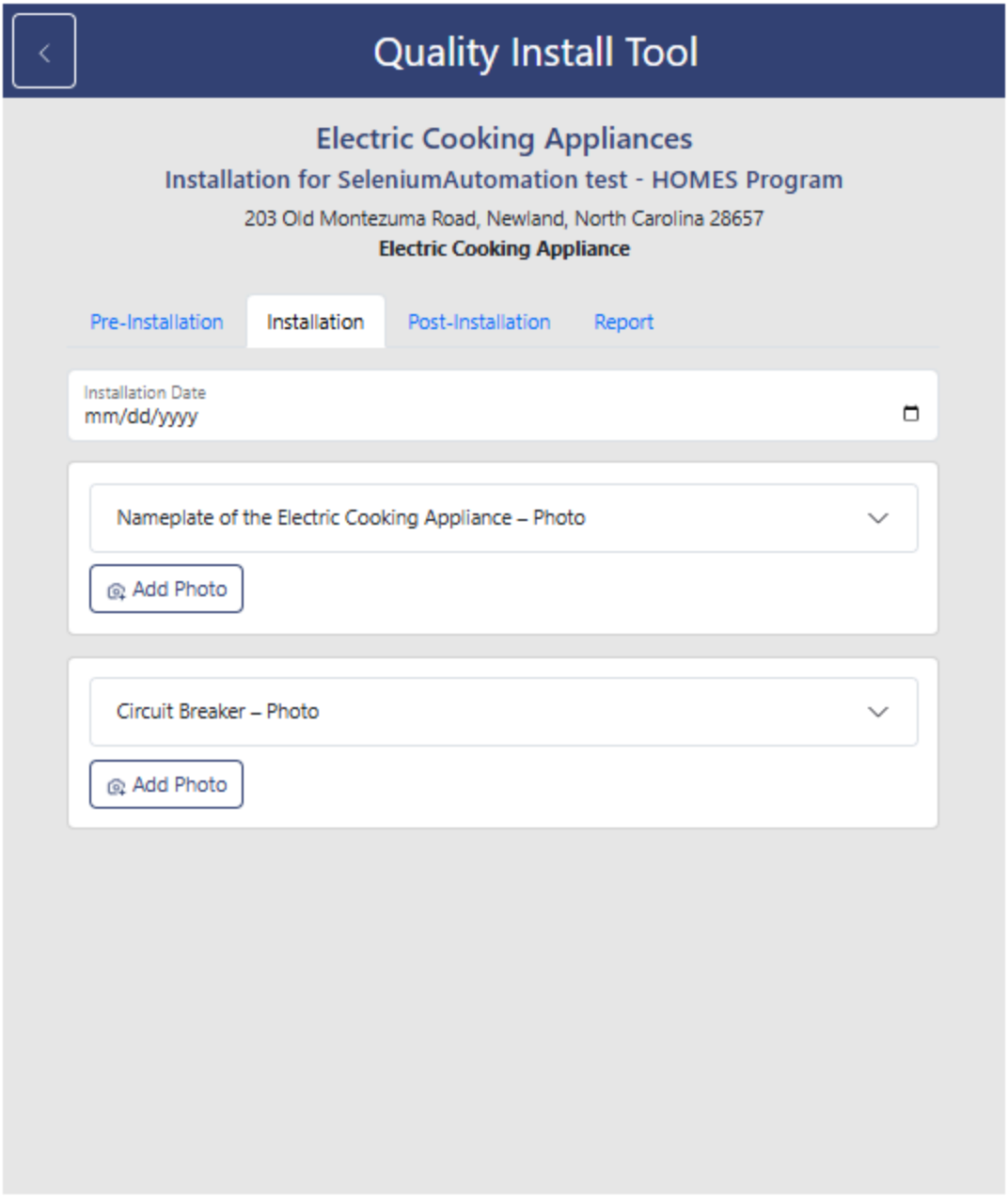
Report tab

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Electric Cooking (HOMES and HEAR)

Add installation

Enter New installation name



Pre-Installation tab

- What is the fuel type of the existing stovetop/oven/range?
    - Gas
    - Electric
    - None

Installation tab

- Installation Date Not required
- PHOTO: Nameplate

- PHOTO: Circuit Breaker
- Energy Saver NC also requests the following photos, which can be uploaded in the fields above. PHOTOS: existing appliance overall and nameplate before removal, gas capping if applicable

Post-Installation tab

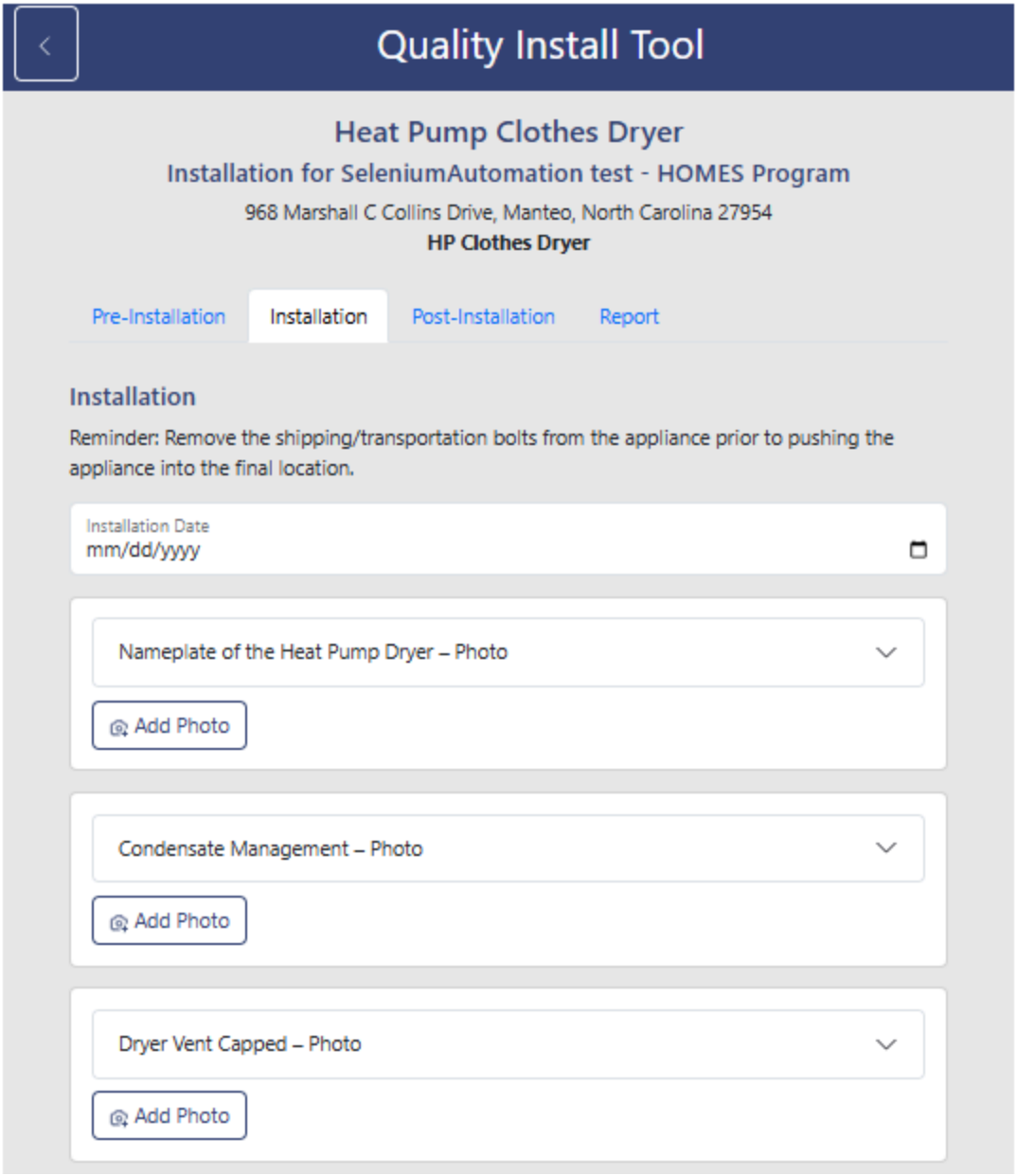
- What is the new stovetop/oven/range type?
    - Standard electric
    - Induction cooktop

Report tab

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Heat Pump Clothes Dryer (HOMES and HEAR)

Add installation
Enter New installation name



Pre-Installation tab

- What is the existing dryer fuel type?
    - Gas
    - Electric
    - None

Installation tab

- Installation Date Not required
- PHOTO: Nameplate
- PHOTO: Condensate drain (drainpipe and drain) Not required

- PHOTO: Dryer vent capped/air sealed
- PHOTO: Dryer circuit breaker Not required
- Energy Saver NC also requests the following photos, which can be uploaded in the fields above. PHOTO: existing appliance overall and nameplate before removal, gas capping if applicable

Post-Installation tab

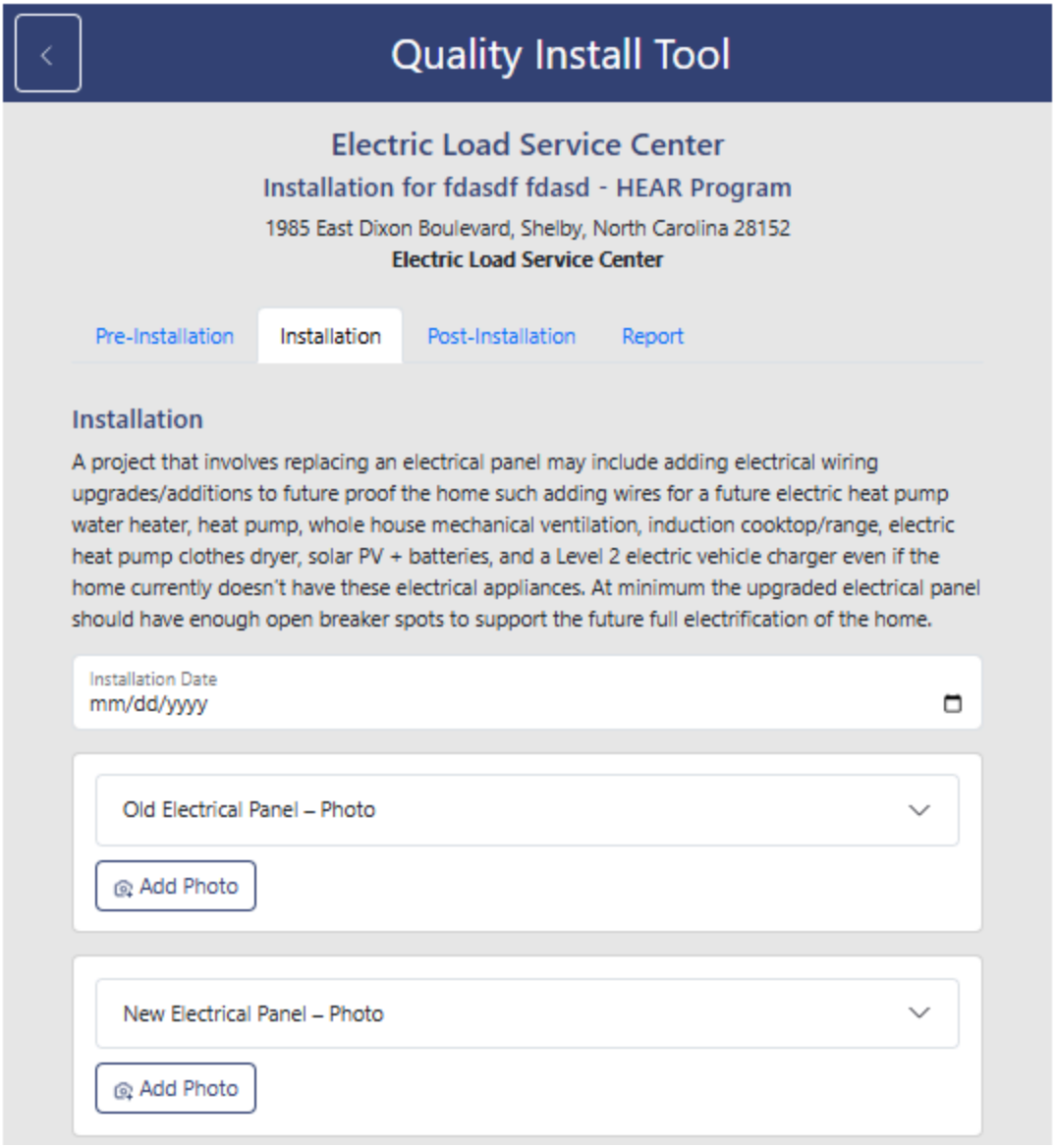
- What is the annual energy use estimated on the EnergyGuide label of the new clothes dryer? (kWh/yr)

Report tab

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Electric Panel Upgrade (HEAR only)

Add installation
Enter New installation name



Pre-Installation tab

- What is the existing electrical panel's max amps to the home?
- How many open breaker spots are there in the existing electrical panel?

Installation tab

For the pre- and post-install photos of the electrical panel, please take the photos with the panel cover removed so it's easy to identify any circuit breakers that were touched in the wiring process.

- Installation Date Not required
- PHOTO: Old panel

- PHOTO: New panel (show amperage) and service entry to house if service is upgraded

Post-Installation tab

- What is the new electrical panel's max amps to the home? (200 minimum recommended or smart panel)
- How many open breaker spots are there in the new electrical panel? (enough to support future electrification of the home)

Report tab

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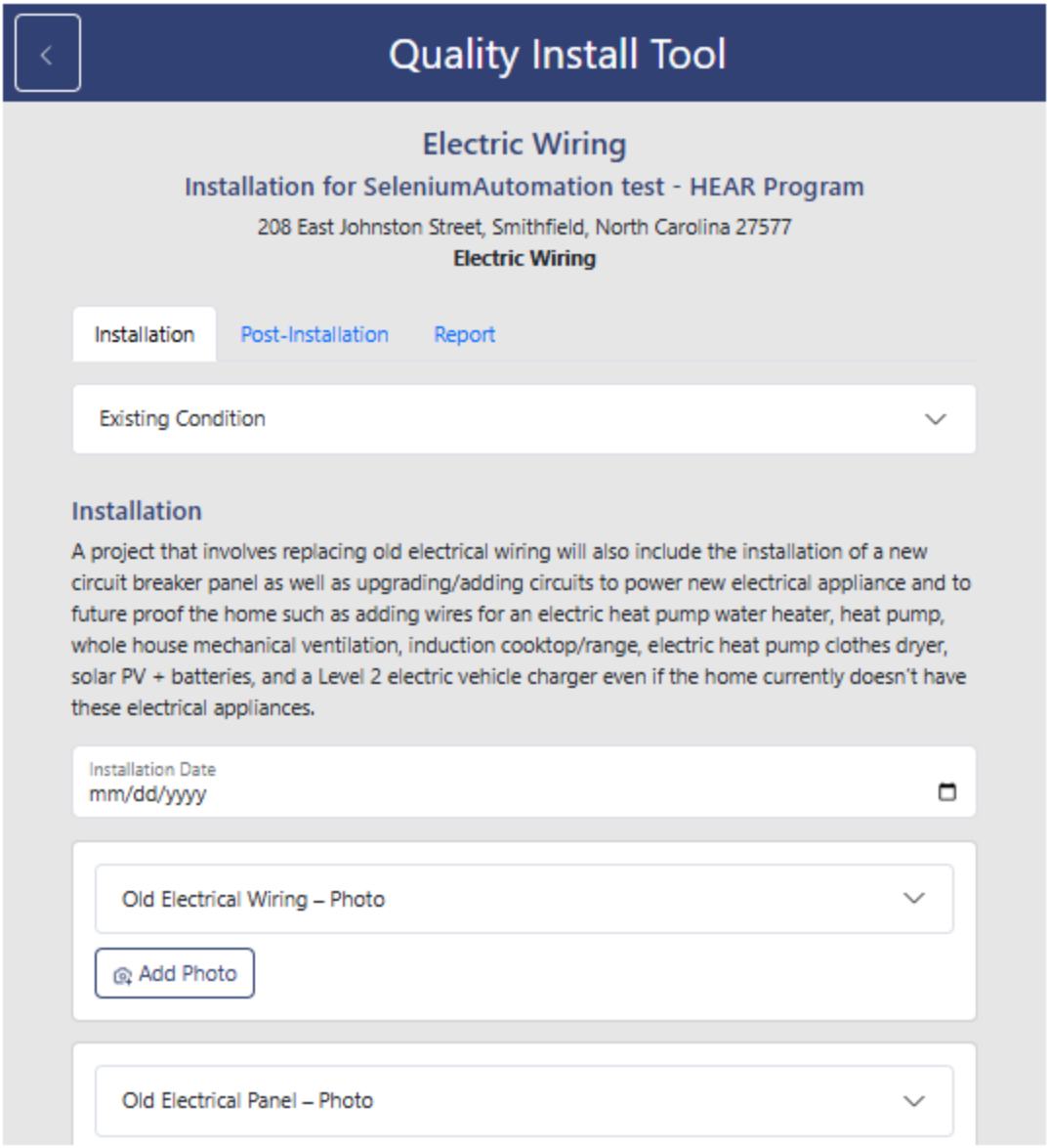
Electric Wiring Upgrade (HEAR Only)

The circuitry plan only needs to be for the affected areas, not the whole house unless you are rewiring the whole house. The circuitry plan can be hand drawn as long as it's clear and legible.

Note that while the electric wiring workflow asks questions about the panel, if you are upgrading the electric panel itself or installing smart load devices which are part of the separate HEAR program electric panel upgrade rebate, please use the separate Electric Load Service Center workflow to document the panel change.

Add installation

Enter New installation name



Installation tab

- Installation Date Not required
- PHOTO: Existing wiring and breakers, if applicable
- PHOTO: Panel prior to installation (cover removed)
- PHOTO: Existing circuitry plan Not required
- PHOTO: New circuitry plan Not required
- PHOTO: New wires and breakers
- PHOTO: Panel post installation (cover removed)

Post-Installation tab

- New panel max amps (200 minimum) Not required

- How many open breaker spots are there in the new panel Not required

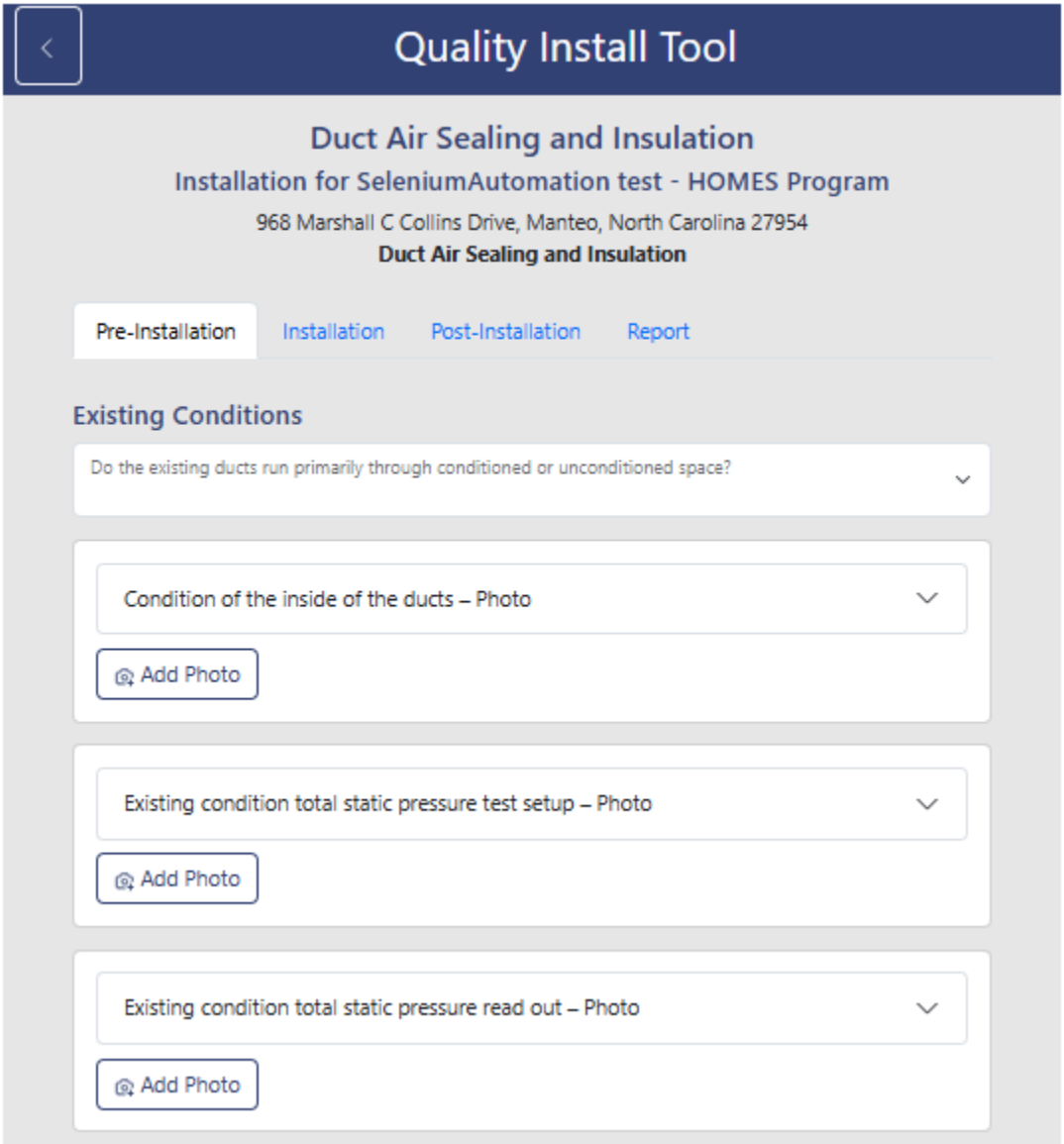
Report tab

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Duct Air Sealing and Insulation (HOMES and HEAR)

Add installation

Enter New installation name



Pre-Installation tab

Conducting a static pressure test is not required, but will help ensure that there aren't other problems needing correction other than sealing ducts. While the recommended test is total static pressure, it's recommended that you also isolate the duct pressure by taking a measurement in the supply plenum after the coil (if equipped). See Use Static Pressure Measurement to Pinpoint Duct Deficiencies - ACCA HVAC Blog for more details. While you are not required to fix undersized ducts, it's recommended to present any deficiencies to your customer and if they upgrade any duct work, then seal the new duct work instead of the old. If the pressure is much too high, sealing the duct work will only make the problem worse and lower system efficiency.

Using a duct blaster to measure total duct leakage pre- and post-installation isn't required, though it is an effective way to prove your results to the customer.

If the customer has any comfort issues isolated to a particular room, you can also take register airflow measurements pre- and post-installation to identify if the air flow to the room was the issue and how the air sealing improved it. Again, like the duct blaster test, this is not a program requirement, but please include the photos if the testing is conducted.

- Do the existing ducts run primarily through conditioned or unconditioned space? Conditioned/Unconditioned
- PHOTO: Condition of the inside of the ducts
- PHOTO: Existing condition total static pressure test setup Not required
- PHOTO: Existing condition total static pressure test result Not required
- What is the total duct leakage before air sealing?
    - CFM at 25Pa
        - PHOTO: Conditioned floor area calculation Not required
        - CFM at 25Pa: Not required
        - Conditioned floor area: Not required
    - Not measured
- PHOTO: Areas of significant leakage
- PHOTO: Undersized ducts, returns or supplies

Installation tab

Take pictures of all areas where sealing is performed. If you can capture multiple duct to duct connections in one photo, that's fine, but you can also provide multiple photos per section.

Pay particular attention to fixing all duct leaks near the system, where pressure is the highest, as leaks where there is higher pressure has a much larger effect on duct loss than a small leak at the end of a long run.

- Installation Date Not required
- PHOTO: Secure and seal duct boots
- PHOTO: Duct to duct connection sealed
- PHOTO: Duct insulation and vapor barrier (show R-value marking)
- PHOTO: Duct support
- PHOTO: Take off collar air sealed
- PHOTO: Plenum seams air sealed
- PHOTO: Air handler access panel seams sealed
- PHOTO: Air filter grille/slot sealed

Post-Installation tab

Conduct the same total static pressure test as pre-install, which should show an improvement in pressure, but not too high of a pressure to cause the system to strain and run inefficiently. If there were any comfort problems associated with particular rooms and you tested the airflow pre-installation, provide photos of the same register airflows tested post-installation.

- What percentage of the duct system received additional insulation?
    - 10%
    - 25%
    - 50%
    - 75%
    - 100%
- PHOTO: Post installation total static pressure test setup Not required
- PHOTO: Post installation total static pressure test result Not required
- What is the total duct leakage before air sealing? Not required
    - CFM at 25Pa
        - CFM at 25Pa:
- PHOTO: Flow rates to each supply register and zonal pressures post-work measurement Not required

Report tab

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Insulation, Air Sealing, Ventilation (HOMES and HEAR)

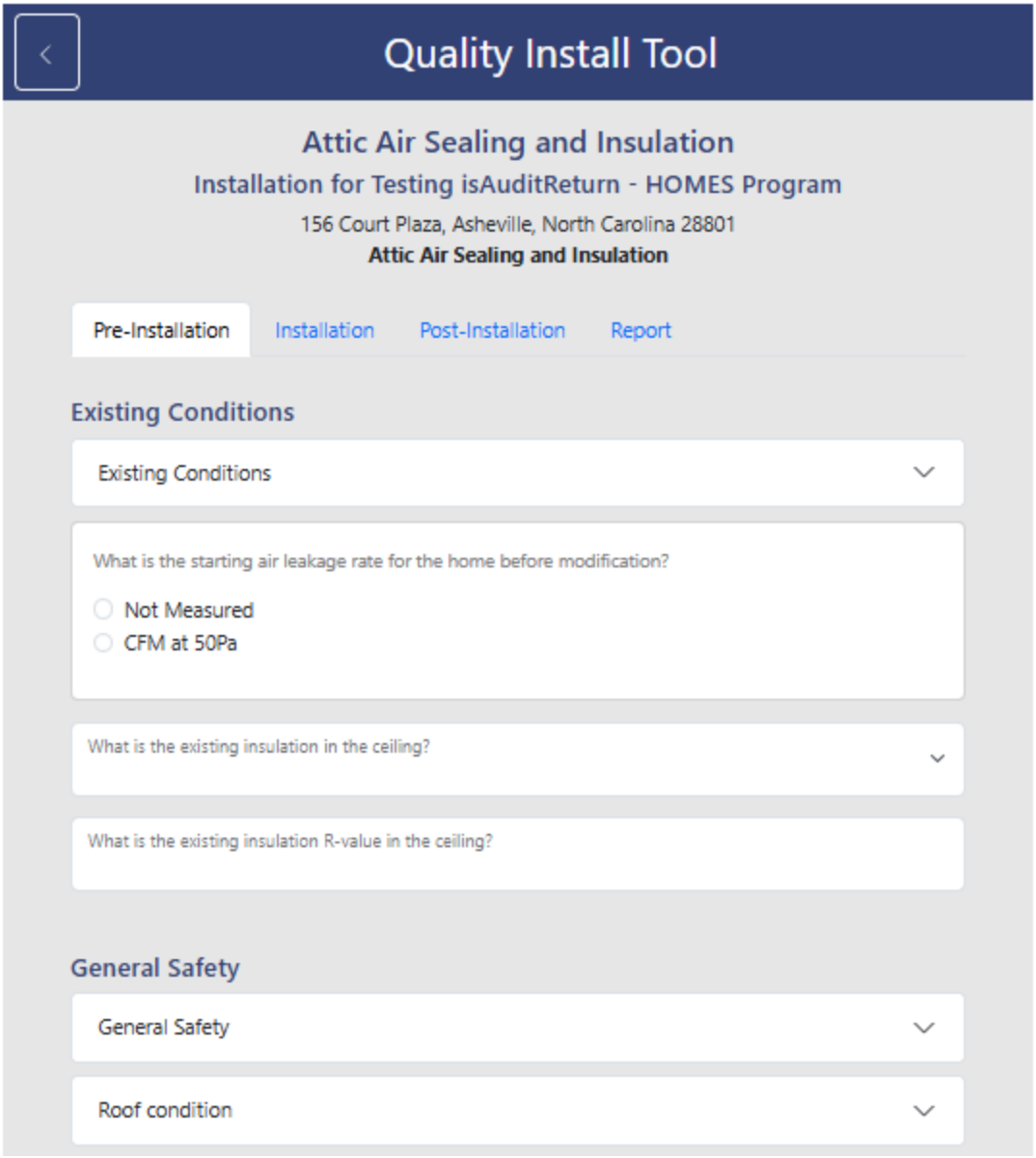
Choose an installation Workflow from:

- Attic Air Sealing and Insulation
- Floor Air Sealing and Insulation above Unconditioned Space
- Foundation Wall Air Sealing and Insulation
- Mechanical Ventilation
- Wall Air Sealing and Insulation (Drill and Fill)

Attic Air Sealing and Insulation

Add installation

Enter New installation name



Pre-Installation tab

- Existing Conditions
    - What is the Starting air leakage rate for the home before modification?
        - Not measured
        - CFM at 50Pa
            - CFM at 50Pa: Not required
            - Conditioned Volume: Not required
            - PHOTO: Starting blower door manometer showing the CFM50 value Not required

            - PHOTO: Conditioned floor area and air volume calculation Not required
        - What is the existing insulation in the ceiling?
            - None
            - Batt insulation
            - Blown insulation
            - Rigid board insulation
            - Closed cell spray foam insulation
            - Other
        - What is the existing insulation R-value in the ceiling?
- General Safety
    - What is the roof condition?
        - Good
        - Potential Issue
    - Are there signs of mold or moisture problems in the attic? Yes/No
    - Has mechanical ventilation calculation been performed pre-retrofit? Yes/No
    - Have the answers to these safety questions been disclosed to the homeowner? Yes/No

Installation tab

Please provide applicable photos and comments if there is something difficult to see or understand from the photo. Remember, multiple photos can be added in each section to accurately reflect your work. Please provide at least one photo for each applicable item below. Before and after photos are always appreciated and will expedite the process.

- Installation Date Not required

  Air Sealing sub-tab
    - PHOTO: Drywall to top plate seam sealed
    - PHOTO: Soffit air baffles installed, blocked and/or air sealed
    - PHOTO: Leaky recessed light treatment
    - PHOTO: Chimney dam, sheet metal and fire caulking
    - PHOTO: Hot flue pipe dam, sheet metal and fire caulking
    - PHOTO: Electrical wire and pipe penetrations sealed
    - PHOTO: Large openings sealed (i.e. dropped soffits, chases)

- PHOTO: Duct boot to drywall sealed
- PHOTO: Exhaust fan to drywall sealed
- PHOTO: Knee walls sealed (top/bottom plates, and back of wall sealed)
- PHOTO: Knee walls enclosed (top/bottom plates, front and back of wall installed)
- PHOTO: Skylight walls sealed (top/bottom plates, and back of wall sealed)
- PHOTO: Skylight walls enclosed (top/bottom plates, front and back of wall installed)
- PHOTO: Balloon walls sealed (blocking and spray foam)
- PHOTO: Stairwell wall sealed
- PHOTO: T&G ceiling sealed
- PHOTO: Storage platform prep (area underneath platform prepped to dense packed with blown insulation)
- PHOTO: Overall of air sealed attic
- PHOTO: Attic access dam (installed)
- PHOTO: Attic access weatherstripped

Insulation sub-tab

- PHOTO: Insulation measuring rulers (installed with insulation)
- PHOTO: R-value per inch table from insulation bag
- PHOTO: Skylight shaft insulated
- PHOTO: Knee wall insulated
- PHOTO: Stair wall insulated
- PHOTO: Storage Platform insulated
- PHOTO: Overall of insulated attic (showing insulation depth)
- PHOTO: Attic access insulated (rigid foam board recommended)

Post-Installation tab

Enter the applicable data. For homes that were sealed, if they have atmospherically vented gas appliances, safety testing is required.

- What is the air sealing level you have achieved?
  - Not measured
  - CFM at 50Pa

- Is the homeowner aware that combustion safety tests are highly recommended for homes with any atmospherically drafted gas appliances after air sealing projects are completed?
    - Yes/No/NA
- What R-value was added to the ceiling?
    - Less than R-30
    - R-40
    - R-49
    - R-60
    - More than R-60
- What percentage of ceiling area received additional insulation?
    - 10%
    - 25%
    - 50%
    - 75%
    - 100%
- Combustion Appliance Safety Testing (Do not report here, reported in Closeout step)
    - If the home upgrades will result in a barometric draft controlled or Category I appliance equipped with a draft hood or connected to a natural draft venting system remaining in the conditioned envelope of the home, combustion safety checks are required to be performed at the completion of the project.
- Has a mechanical ventilation calculation been performed post-retrofit to ensure proper indoor air quality? Yes/No
- Project Invoice Not required (uploaded on Closeout tab)
    - PHOTO: Project Invoice

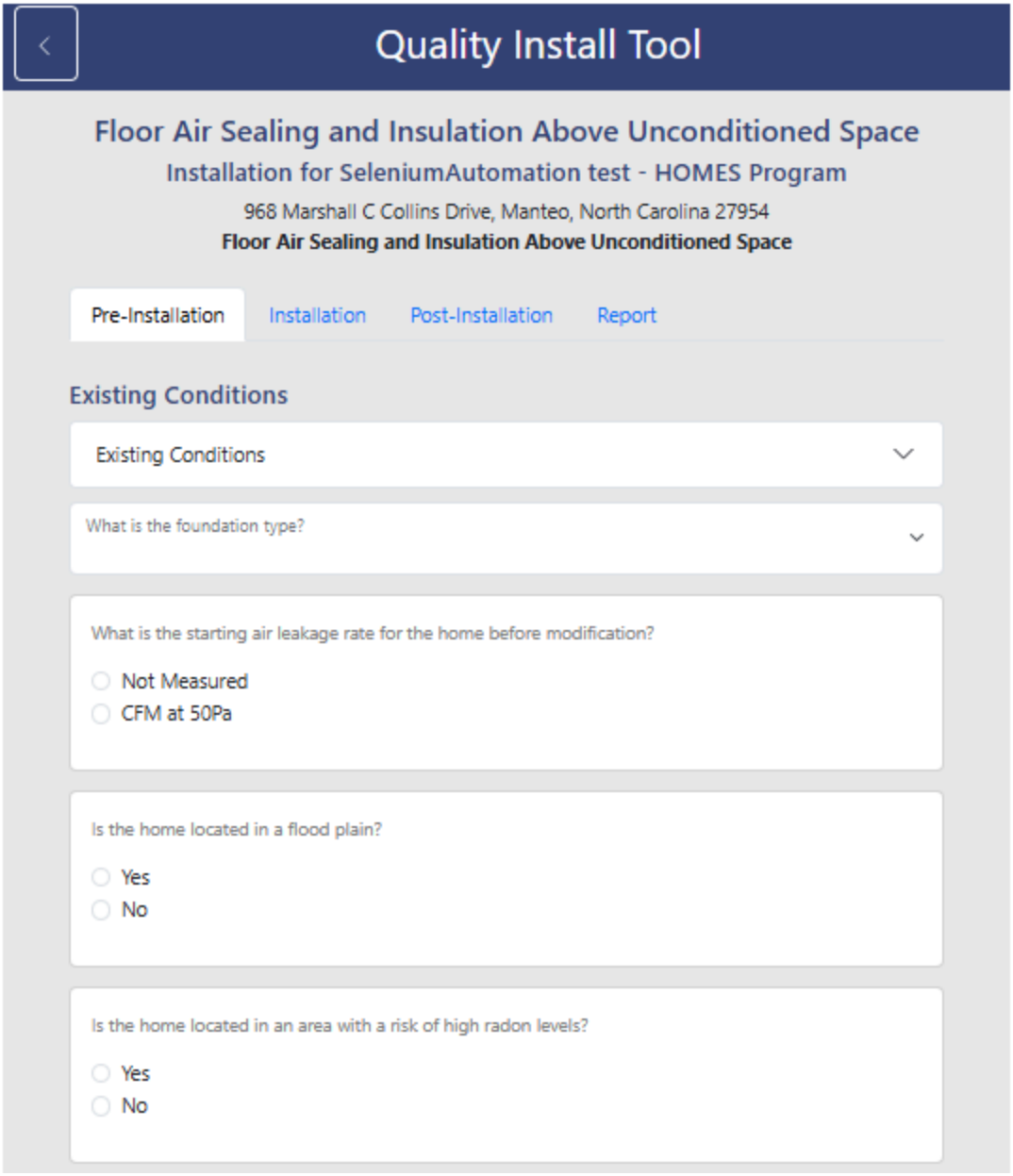
Report

This tab allows you to see all inputs from other tabs, Need to hit Submit Final Report

Floor Air Sealing and Insulation above Unconditioned Space

Add installation

Enter New installation name



Pre-Installation tab

- What is the foundation type?
    - Vented crawlspace
    - Unconditioned basement
- What is the starting air leakage rate for the home before modification?
    - Not measured
    - CFM50
        - CFM at 50 Pa
        - Conditioned volume

        - PHOTO: Starting blower door test result Not required
        - PHOTO: Conditioned floor area and air volume calculation (math and sketch) Not required
- Is the home located in a flood plain? Yes/No
- Is the home located in an area with a risk of high radon levels? Yes/No
- What percentage of floor area is above unconditioned space?
    - 10%
    - 25%
    - 50%
    - 75%
    - 100%
- What is the drainage system condition?
    - Good
    - Potential issue
- Are there signs of mold or moisture in or outside the home? Yes/No
- Has ASHRAE 62.2 calculation been performed pre-retrofit? Yes/No
- Have the answers to these safety questions been disclosed to the homeowner? Yes/No

Installation tab

Before and after photos are always appreciated and will expedite the process.

- Installation date Not required
- PHOTO: Penetrations through subfloor sealed
- PHOTO: Floor joists air sealed and insulated
- PHOTO: Foam board penetrations sealed
- PHOTO: Foundation floor (sealed vapor barrier or concrete)
- PHOTO: Crawl space vents with pest resistant screens
- PHOTO: Crawl space door/access

Post-Installation tab

- What R-value was added to the floor?
    - Less than R-10
    - R-10
    - R-13
    - R-19
    - R-30
    - More than R-30

- What is the air sealing level you have achieved?
    - Not measured
    - CFM at 50Pa
        - CFM at 50Pa
        - PHOTO: Final blower door test result Not required
- Combustion Appliance Safety Testing (Do not report here, reported in Closeout step)
    - If the home upgrades will result in a barometric draft controlled or Category I appliance equipped with a draft hood or connected to a natural draft venting system remaining in the conditioned envelope of the home, combustion safety checks are required to be performed at the completion of the project.
- Has an ASHRAE 62.2 mechanical ventilation calculation been performed post-retrofit to ensure proper indoor air quality? Yes/No
- Project Invoice Not required (uploaded on Closeout tab)
    - PHOTO: Project Invoice

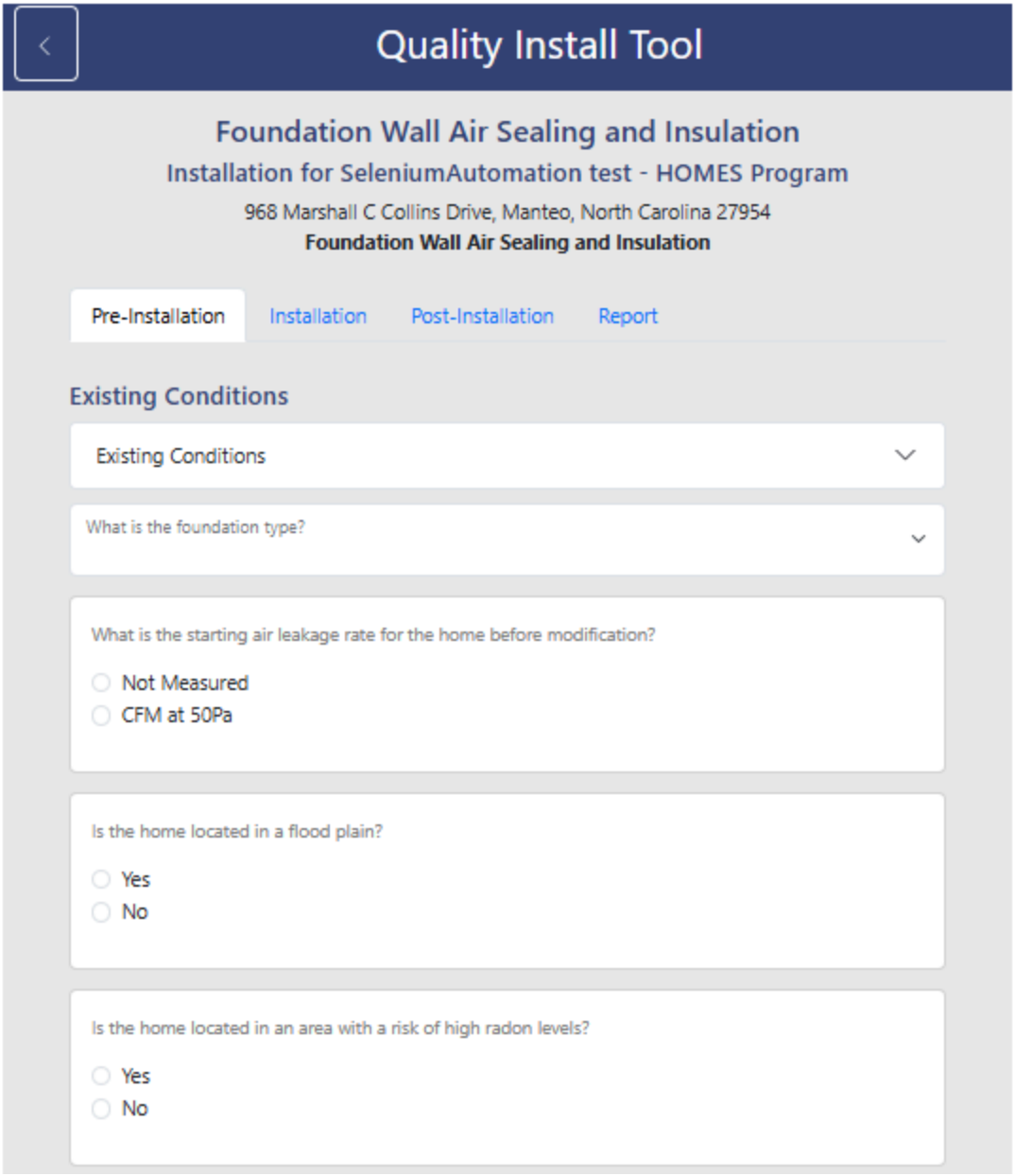
Report tab

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Foundation Wall Air Sealing and Insulation

Add installation

Enter New installation name



Pre-Installation tab

- What is the foundation type?
    - Conditioned crawl space
    - Unconditioned crawl space to be encapsulated
    - Conditioned basement
- What is the starting air leakage rate for the home before modification?
    - Not Measured
    - CFM at 50Pa
        - CFM at 50Pa

        - Conditioned Volume
        - PHOTO: Starting blower door test result Not required
        - PHOTO: Conditioned floor area and air volume calculation (math and sketch) Not required
- Is the home located in a flood plain? Yes/No
- Is the home located in an area with a risk of high radon levels? Yes/No
- What is the material of the foundation wall?
    - Wood
    - Concrete/CMU
- What is the drainage system condition?
    - Good
    - Potential issue
- Are there signs of mold or moisture in or outside the home? Yes/No
- Has ASHRAE 62.2 calculation been performed pre-retrofit? Yes/No
- Have the answers to these safety questions been disclosed to the homeowner? Yes/No

Installation tab

Before and after photos are always appreciated and will expedite the process.

- Installation date Not required
- PHOTO: Overall photo(s) of existing conditions
- PHOTO: Moisture test photo Not required
- PHOTO: Rim joist air sealed
- PHOTO: Crawlspace vents sealed
- PHOTO: Foundation Wall Air Sealed
- PHOTO: Foundation Floor (vapor barrier or slab)
- PHOTO: Foundation Wall Insulation

Post-Installation tab

- What percent of the foundation perimeter was insulated?
    - 10%
    - 25%
    - 50%
    - 75%
    - 100%
- What R-value was added to the foundation perimeter?
    - Less than R-10

    - R-10
    - R-13
    - R-19
    - R-30
    - More than R-30
- What is the post air sealing and insulation air leakage rate for the home?
    - Not measured
    - CFM50
        - CFM at 50 Pa
        - PHOTO: Final blower door test result Not required
- Is the homeowner aware that combustion safety tests are highly recommended for homes with any gas appliances after air sealing projects are completed? Yes/No/NA
- Combustion Appliance Safety Testing (Do not report here, reported in Closeout step)
    - If the home upgrades will result in a barometric draft controlled or Category I appliance equipped with a draft hood or connected to a natural draft venting system remaining in the conditioned envelope of the home, combustion safety checks are required to be performed at the completion of the project.
- Has an ASHRAE 62.2 mechanical ventilation calculation been performed post-retrofit to ensure proper indoor air quality? Yes/No
- Project Invoice Not required (uploaded on Closeout tab)
    - PHOTO: Project Invoice

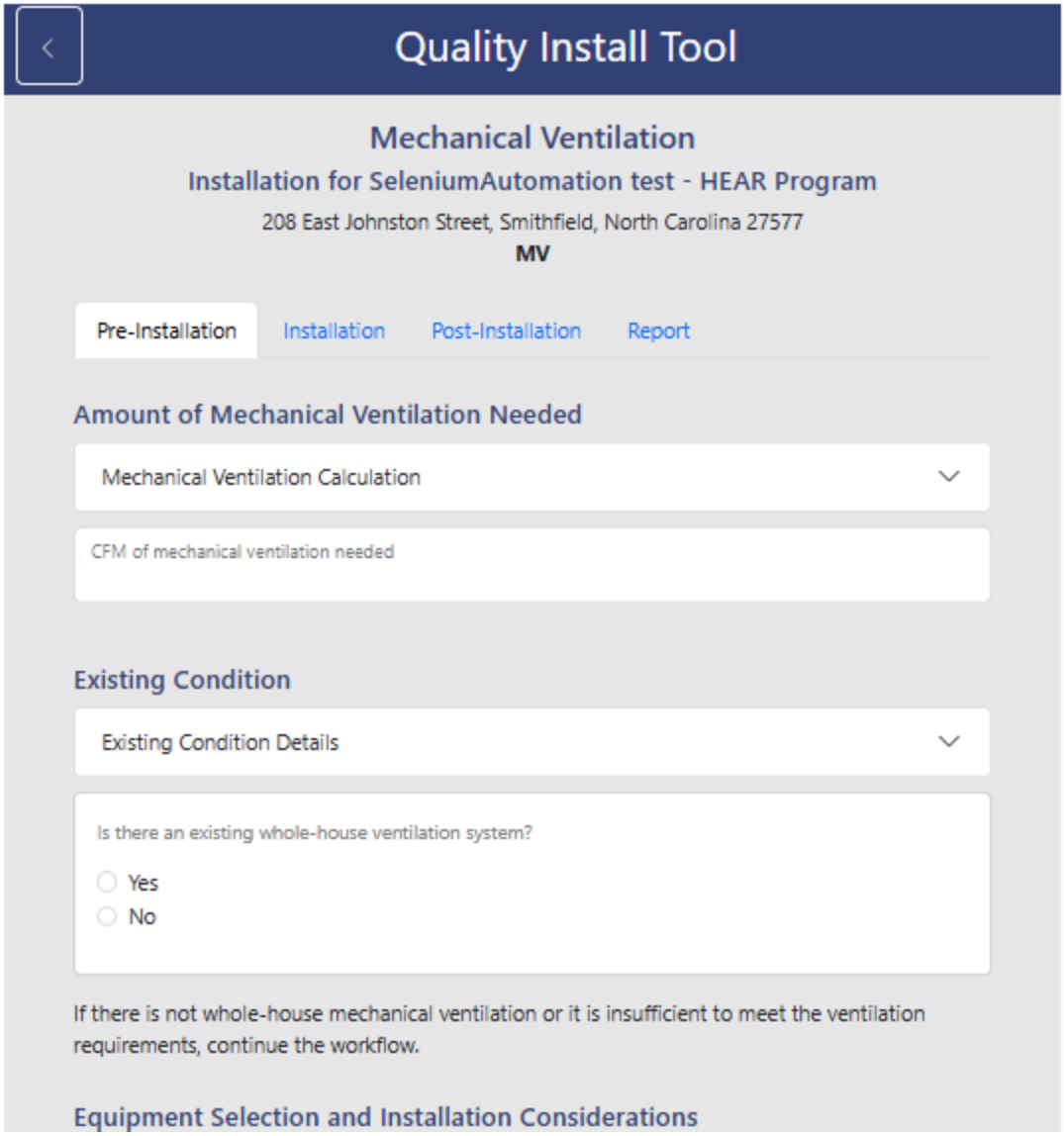
Report tab

This tab allows you to see all inputs from other tabs. If all information is correct and complete, click Submit Final Report. Once you submit your report, your entries cannot be accessed or changed. **If your application is returned, you will have to resubmit all information within the QIT.**

Mechanical Ventilation

Add installation

Enter New installation name



Pre-Installation tab

- CFM of mechanical ventilation needed
- Is there a whole house ventilation system? Yes/No
- Equipment selection and installation considerations
    - Continuous exhaust
    - Supply only
    - Balanced supply and return
    - Energy recovery ventilation
    - Heat recovery ventilation

Installation tab

- Installation date Not required

- PHOTO: Nameplate
- PHOTO: Intake (should be serviceable and labeled) Not required
- PHOTO: Exhaust
- PHOTO: Air flow measurement (Make sure that you use an average of measured speeds across the ventilation source if you aren't using a hood to measure cfm.) Not required
- PHOTO: Mechanical ventilation unit installed (Overall showing access for maintenance and cleaning)

Post-Installation tab

- Combustion Appliance Safety Testing (Do not report here, reported in Closeout step)
    - If the home upgrades will result in a barometric draft controlled or Category I appliance equipped with a draft hood or connected to a natural draft venting system remaining in the conditioned envelope of the home, combustion safety checks are required to be performed at the completion of the project.
- Project Invoice Not required (uploaded on Closeout tab)
    - PHOTO: Project Invoice

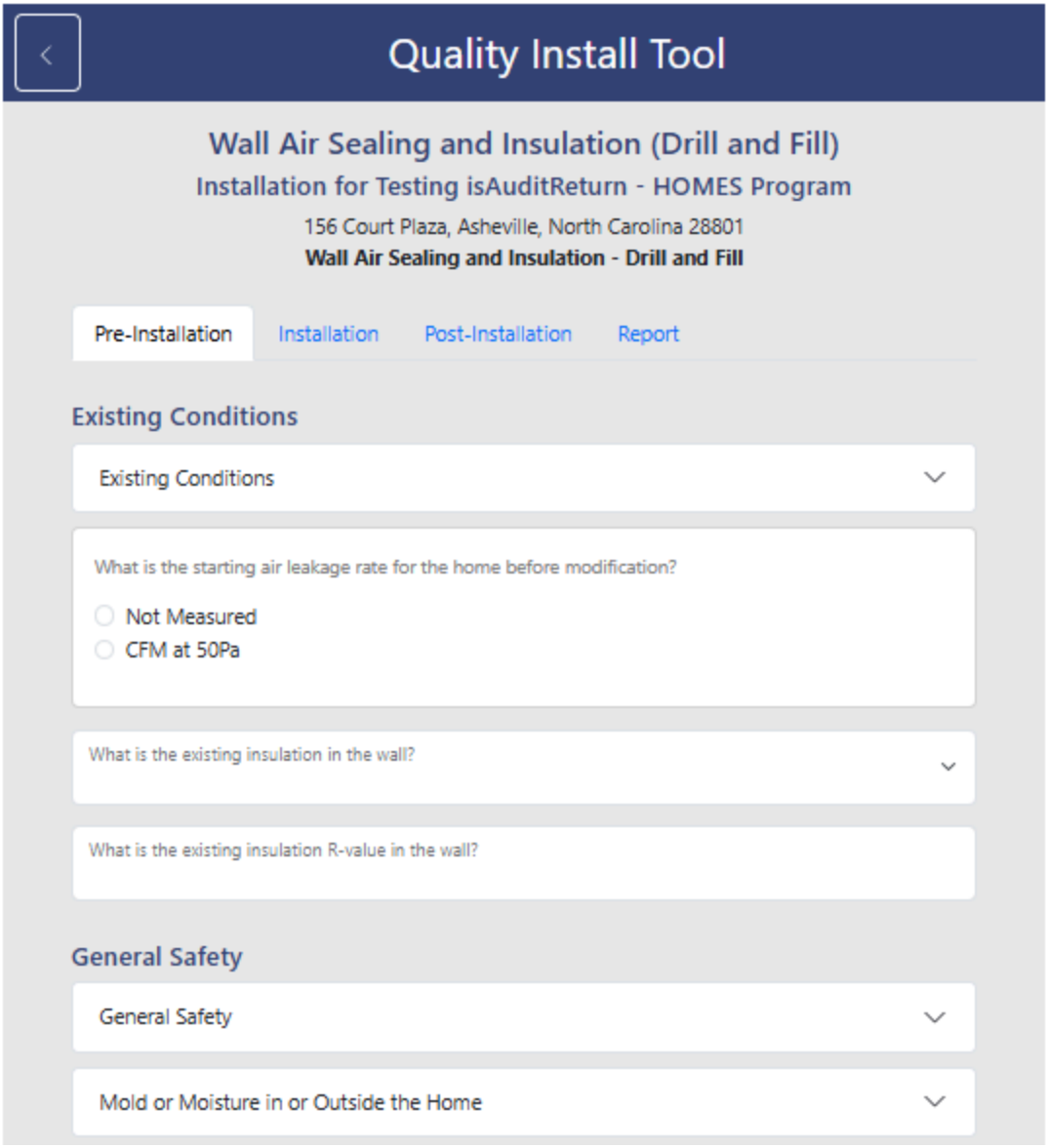
Report tab

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Wall Air Sealing and Insulation (Drill and Fill)

Add installation

Enter New installation name



Pre-Installation tab

- What is the starting air leakage rate for the home before modification?
    - Not Measured
    - CFM at 50Pa
        - CFM at 50Pa
        - Conditioned Volume Not required
        - PHOTO: Starting blower door test result Not required
        - PHOTO: Conditioned floor area and air volume calculation (math and sketch) Not required
- What is the existing insulation in the wall?

    - None
    - Cavity insulation (batt or blown-in)
    - Fanfold insulation
    - Exterior continuous insulation
    - Don't know
- What is the existing insulation R-value in the wall?
- Are there signs of mold or moisture in or outside the home? Yes/No
- Has ASHRAE 62.2 calculation been performed pre-retrofit? Yes/No
- Have the answers to these safety questions been disclosed to the homeowner? Yes/No

Installation tab

Before and after photos are always appreciated and will expedite the process.

- Installation date Not required
- PHOTO: Outlets and switches air sealed
- PHOTO: Windows and doors air sealed
- PHOTO: Rim/band joist between floors air sealed
- PHOTO: Tub/Shower on exterior wall (boroscope)
- PHOTO: Prefabricated metal fireplaces on exterior wall (boroscope or thermal camera, photo of air barrier at exterior wall)
- PHOTO: Chase wall air barrier
- PHOTO: Soffit against exterior wall (exterior wall and ceiling air barrier)
- PHOTO: Through-wall pipe penetration Air Sealed
- PHOTO: Through-wall vent penetration Air Sealed
- PHOTO: Thermal image showing cavities have been filled
- PHOTO: Insulation plugs

Post-Installation tab

- Select one of the Cavity Insulation Improvement:
    - None
    - Cellulose
    - Fiberglass
    - Injection foam
- Resulting R-value of the cavity
    - R-13
    - R-19

    - R-21
    - R-23
    - More than R-23
- Select one of the following Exterior Insulation Improvements:
    - None
    - Exterior continuous insulation - 1 inch
    - Exterior continuous insulation - 2 inches
    - Exterior continuous insulation - more than 2 inches
- What percent of the wall area received additional insulation?
    - 10%
    - 25%
    - 50%
    - 75%
    - 100%
- What is the air sealing level you have achieved?
    - Not measured
    - CFM50
        - CFM at 50 Pa
        - PHOTO: Final blower door test result Not required
- Is the homeowner aware that combustion safety tests are highly recommended for homes with any gas appliances after air sealing projects are completed? Yes/No/NA
- Combustion Appliance Safety Testing (Do not report here, reported in Closeout step)
    - If the home upgrades will result in a barometric draft controlled or Category I appliance equipped with a draft hood or connected to a natural draft venting system remaining in the conditioned envelope of the home, combustion safety checks are required to be performed at the completion of the project.
- Has an ASHRAE 62.2 mechanical ventilation calculation been performed post-retrofit to ensure proper indoor air quality? Yes/No
- Project Invoice Not required (uploaded on Closeout tab)
    - PHOTO: Project Invoice

Report tab

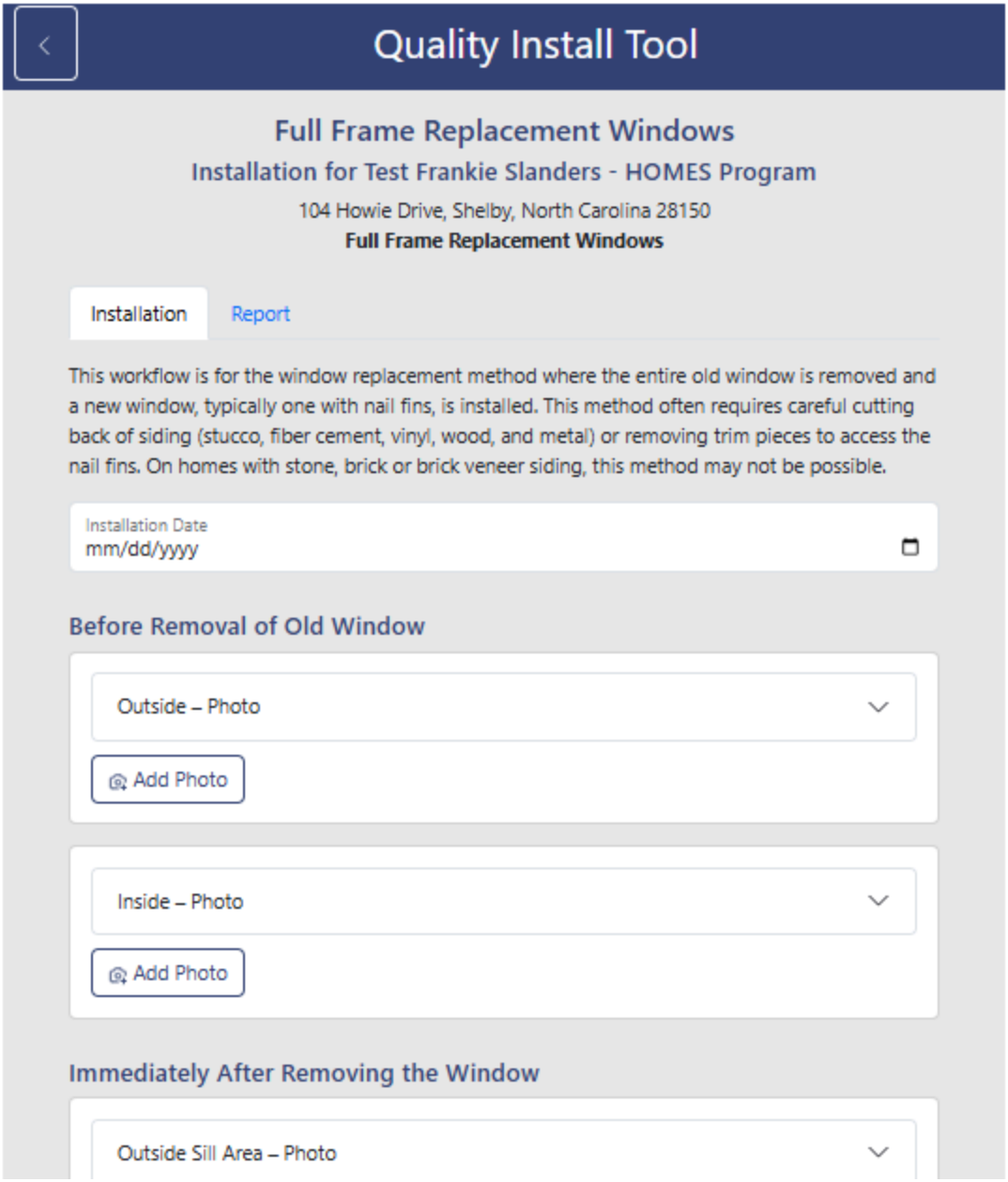
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Full Frame Replacement Windows (HOMES Only)

Photos should be provided for each window replacement. You can upload multiple photos for each section. If there were any repairs to the structure after removing the windows, please add that to the notes as well as anything else that would be important for someone to understand the work performed, particularly if there are any issues visible in the photos, the notes should indicate how they were repaired, such as replacing a rotted framing member or removing and patching a section of rotted wood.

Make sure to document proper air sealing around the window as well before closing it back up.

Add installation
Enter New installation name



Installation tab

If one photo shows all details requested, only one photo upload is required

- Installation Date Not required
- Before Removal of Old Window
    - PHOTO: Original window from outside
    - PHOTO: Original window from inside Not required
- Immediately After Removing the Window
    - PHOTO: Outside sill area
    - PHOTO: Outside head area

- Rough Opening Flashing
    - PHOTO: Outside sill area
    - PHOTO: Outside head area
- After installing the Window
    - PHOTO: Outside sill area (flashing to nail fin relationship)
    - PHOTO: Outside head area (flashing to nail fin relationship)
    - PHOTO: Inside rough opening air sealing
- Completion Photos
    - PHOTO: New window from outside (siding and trim have been repaired/reinstalled and caulked)
    - PHOTO: New window from inside (trim and drywall have been repaired/reinstalled and caulked)
    - Energy Saver NC also requests the following photo(s), which can be uploaded in the fields above, if already visible in photos above it is not required. PHOTO: NFRC label clearly showing U-value and SHGC in one of the fields below.

Report tab

This tab allows you to see all inputs from other tabs. If all information is correct and complete, click Submit Final Report. Once you submit your report, your entries cannot be accessed or changed. **If your application is returned, you will have to resubmit all information within the QIT.**

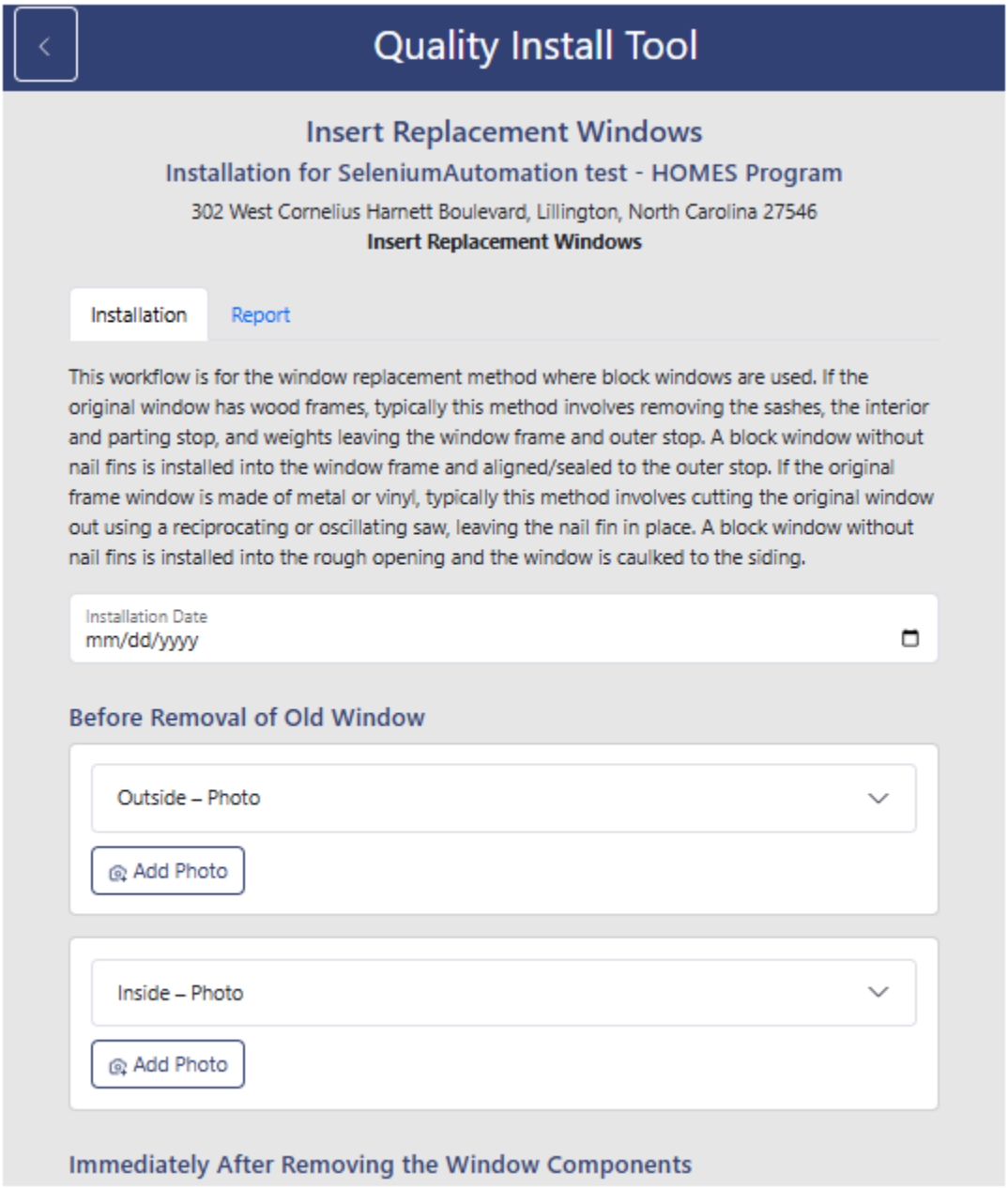
Insert Replacement Windows (HOMES Only)

Photos should be provided for each window replacement. You can upload multiple photos for each section. If there were any repairs to the structure after removing the windows, please add that to the notes as well as anything else that would be important for someone to understand the work performed, particularly if there are any issues visible in the photos, the notes should indicate how they were repaired, such as replacing a rotted framing member or removing and patching a section of rotted wood.

Make sure to document proper air sealing around the window as well before closing it back up.

Add installation

Enter New installation name



Installation tab

If one photo shows all details requested, only one photo upload is required

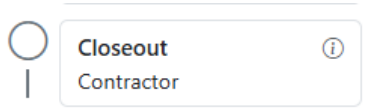
- Installation Date Not required
- Before Removal of Old Window
    - PHOTO: Original window from outside
    - PHOTO: Original window from inside Not required
- Immediately After Removing the Window

    - PHOTO: Outside sill area
    - PHOTO: Outside head area
- Rough Opening Flashing
    - PHOTO: Outside sill area
    - PHOTO: Outside head area
- After installing the Window
    - PHOTO: Outside sill area (flashing to nail fin relationship)
    - PHOTO: Outside head area (flashing to nail fin relationship)
    - PHOTO: Inside rough opening air sealing
- Completion Photos
    - PHOTO: New window from outside (siding and trim have been repaired/reinstalled and caulked)
    - PHOTO: New window from inside (trim and drywall have been repaired/reinstalled and caulked)
    - Energy Saver NC also requests the following photo(s), which can be uploaded in the fields above, if already visible in photos above it is not required. PHOTO: NFRC label clearly showing U-value and SHGC in one of the fields below.

Report tab

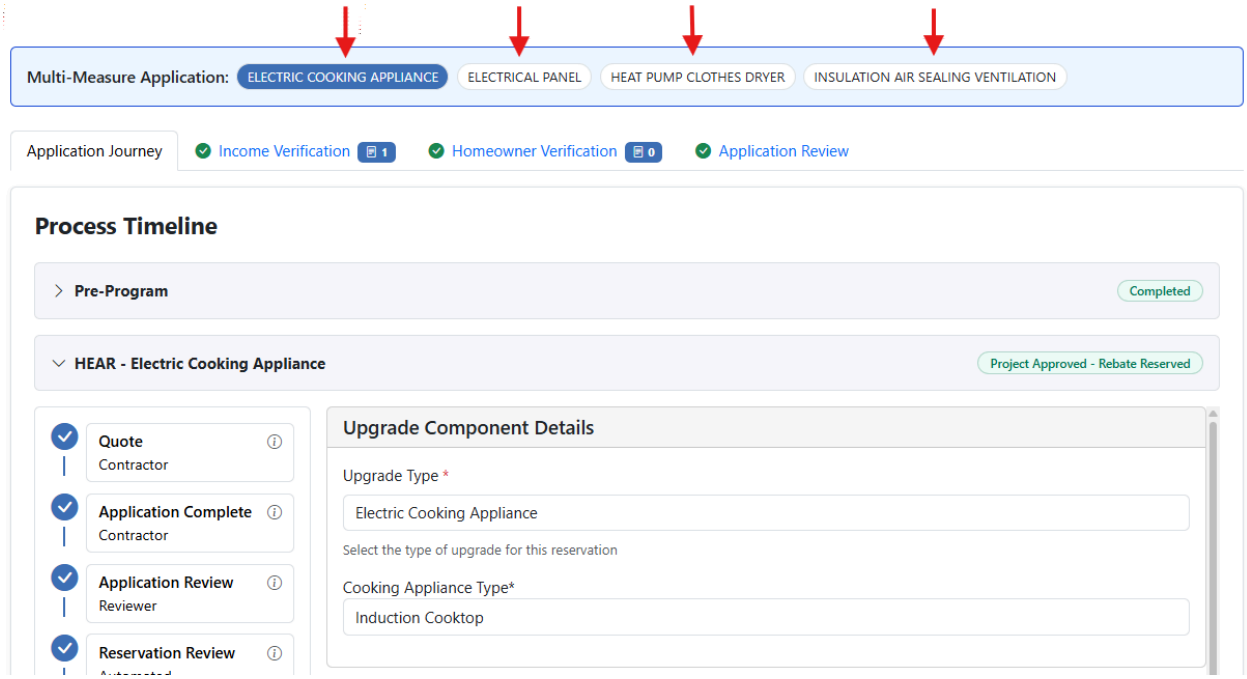
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Closeout step



This step contains sections called Upgrade Component Details that pertain only to HEAR rebates as well as the Safety Checklist, Invoice and Project Summary sections which are required for all rebates.

The Upgrade Component Details for HEAR rebates will only populate if they are a reserved measure available for close-out. You will need to select the appropriate button from the Multi-Measure Application list and then fill in that component's details in the fields provided. You will have to do this for each measure that has a button.



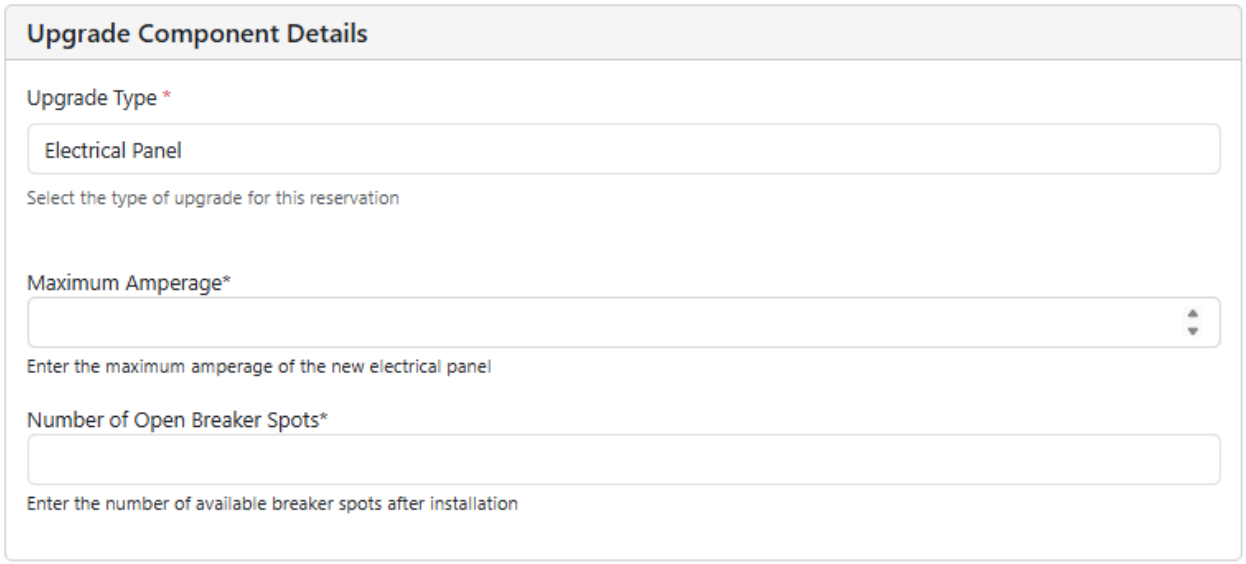
HEAR only:
Upgrade Component Details

- Upgrade Type: (Dropdown)
  - Electrical Panel
  - Electric Cooking Appliance
  - Electric Wiring
  - Heat Pump Clothes Dryer
  - Heat Pump for Space Heating/Cooling

- Heat Pump Water Heater
- Insulation, Air Sealing & Ventilation

HEAR - Electrical Panel (Electric Load Service Center)

Upgrade only if the old panel cannot support the upgrade being installed. Upgrading the panel should future proof the home ensuring they have the space in the new panel to support any future electrification projects such as heat pump water heater, heat pump space heating and/or cooling, electric clothes dryer and cooking equipment, solar PV, and electric vehicle charging. Ensure at least 3 open breaker spots remain post-upgrade.

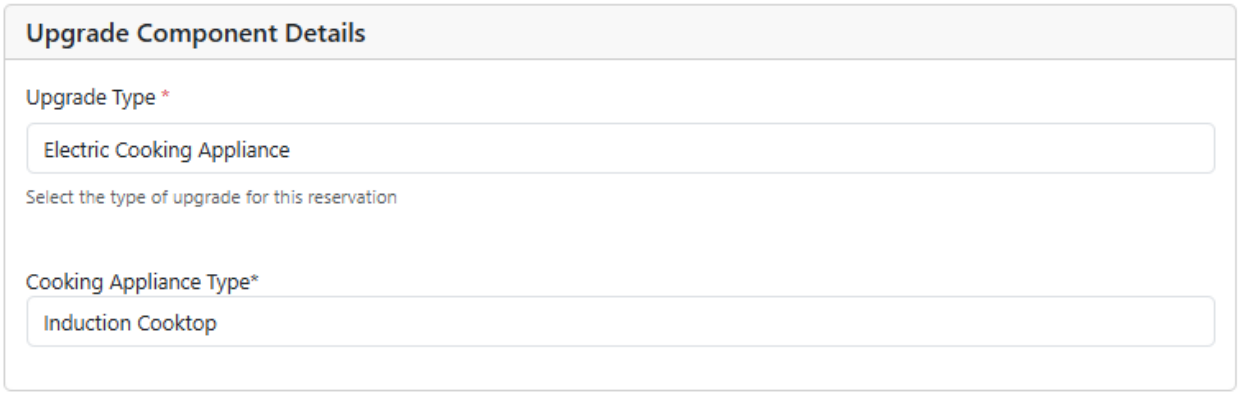


Upgrade Component Details

- Upgrade Type: Electrical Panel
- Maximum Amperage

Number of Open Breaker Spots

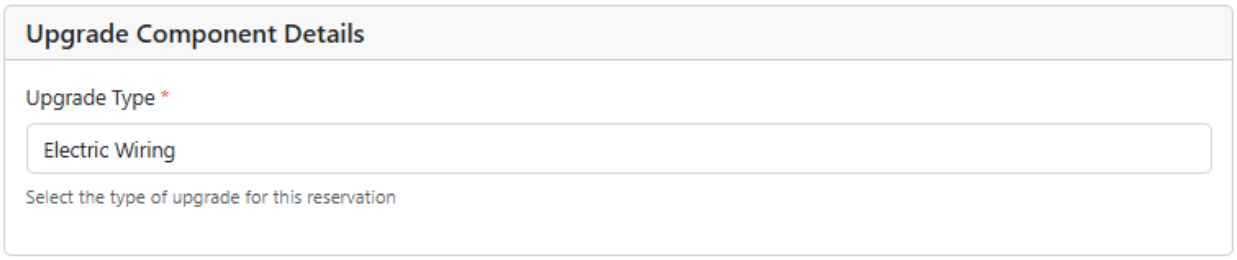
HEAR - Electric Cooking Appliance



Upgrade Component Details

- Upgrade Type: Electric Cooking Appliance
- Cooking Appliance Type
    - Induction Cooktop
    - Standard Electric

HEAR - Electric Wiring



Upgrade Component Details
Upgrade Type: Electric Wiring

HEAR - Heat Pump Clothes Dryer

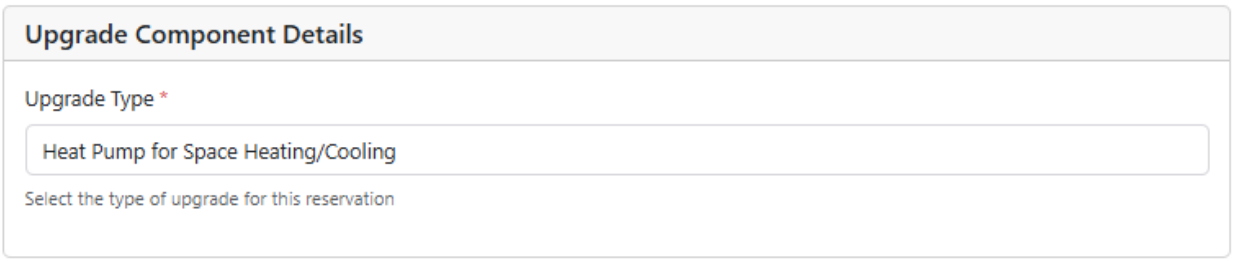


Upgrade Component Details

- Upgrade Type: Heat Pump Clothes Dryer

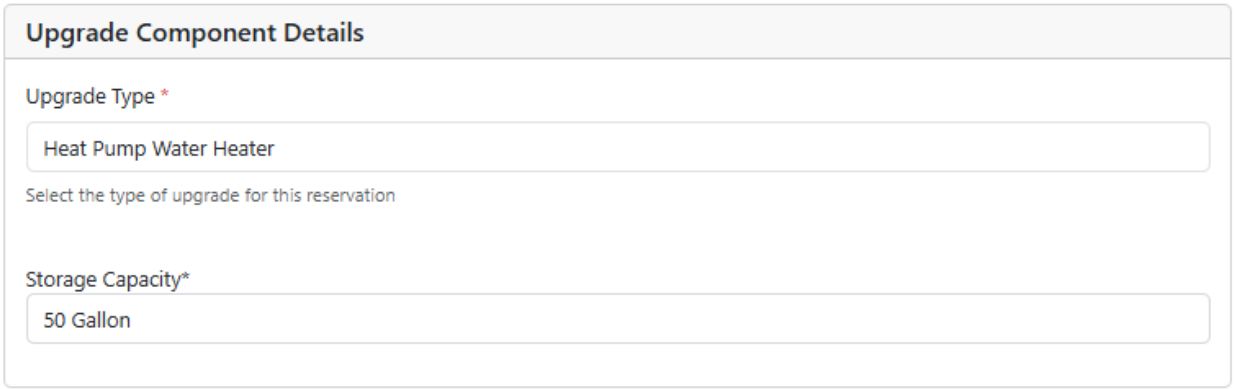
Energy Guide Estimated Energy Use (kWh)

HEAR - Heat Pump for Space Heating or Cooling



Upgrade Component Details
Upgrade Type: Heat Pump for Space Heating/Cooling

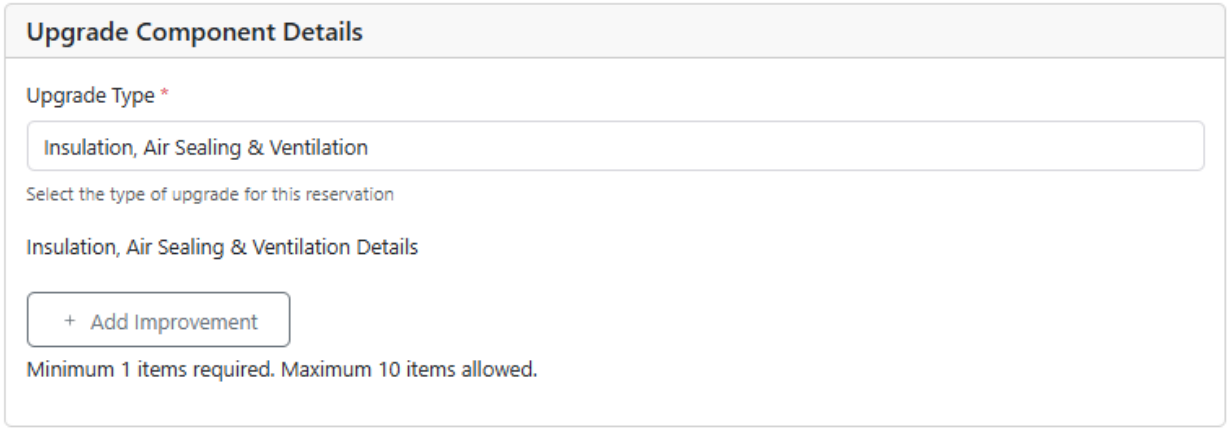
HEAR - Heat Pump Water Heater



Upgrade Component Details

- Upgrade Type: Heat Pump Water Heater
- Storage Capacity (if the exact capacity is not listed here, please choose the closest option)
    - Less than 45 Gallon
    - 45 Gallon
    - 50 Gallon
    - 60 Gallon
    - 80 Gallon
    - More than 80 Gallon

HEAR - Insulation, Air Sealing & Ventilation



Upgrade Component Details

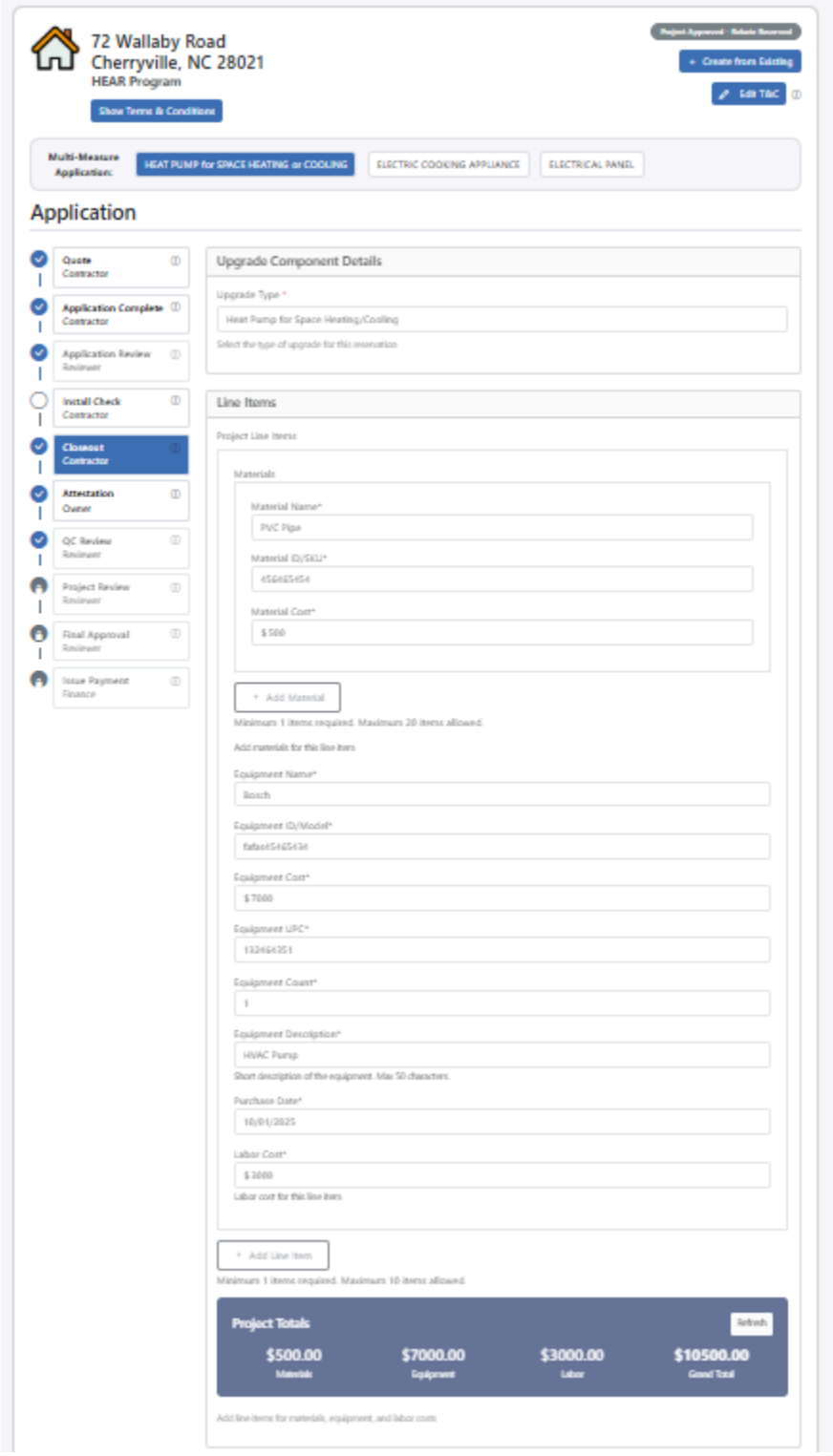
- Upgrade Type: Insulation, Air Sealing & Ventilation
- Insulation, Air Sealing & Ventilation Details, Improvement Type (min 1)
    - Air Sealing
        - Checkbox: Is the leakage rate known? If checked, input CFM50
        - Checkboxes, Locations:
            - Attic
            - Floor/Foundation
            - Walls/Windows/Doors
        - Checkbox: Is the homeowner aware that combustion safety tests are highly recommended? For homes with any combustion appliances after air sealing projects
    - Ceiling Insulation
        - Percent of Ceiling Receiving Additional Insulation
            - 10%
            - 25%

                - 50%
                - 75%
                - 100%
            - R-Value Added
                - Less than R-30
                - R-30
                - R-49
                - R-60
                - More than R-60
    - Wall Insulation
        - Cavity Insulation Added
            - R-13
            - R-19
            - R-21
            - More than R-21
        - Exterior Continuous Insulation Added
            - 1 inch
            - 2 inch
            - More than 2 inch
        - Percent of Wall Area Treated
            - 10%
            - 25%
            - 50%
            - 75%
            - 100%
    - Duct Sealing
        - Checkbox: Is the duct leakage rate known?
    - Duct Insulation
        - Percent of Ducts Receiving Additional Insulation
            - 10%
            - 25%
            - 50%
            - 75%
            - 100%
    - Floor over Conditioned Space Insulation
        - R-Value Added
            - Less than R-30
            - R-30
            - R-49

                - R-60
                - More than R-60
            - Percent of Floor Receiving Additional Insulation
                - 10%
                - 25%
                - 50%
                - 75%
                - 100%
            - R-Value Added (Floor)
                - Less than R-10
                - R-10
                - R-13
                - R-19
                - R-30
                - More than R-30
    - Foundation Insulation
        - R-Value Added
            - Less than R-30
            - R-30
            - R-49
            - R-60
            - More than R-60
        - Percent of Foundation Perimeter Receiving Additional Insulation
            - 10%
            - 25%
            - 50%
            - 75%
            - 100%
        - R-Value Added (Foundation)
            - Less than R-5
            - R-5
            - R-10
            - R-13
            - R-19
            - R-30
    - Ventilation
        - CFM Flow Rate

Line Items (Materials and Equipment)

Fill this section out for all rebates, HOMES and HEAR.



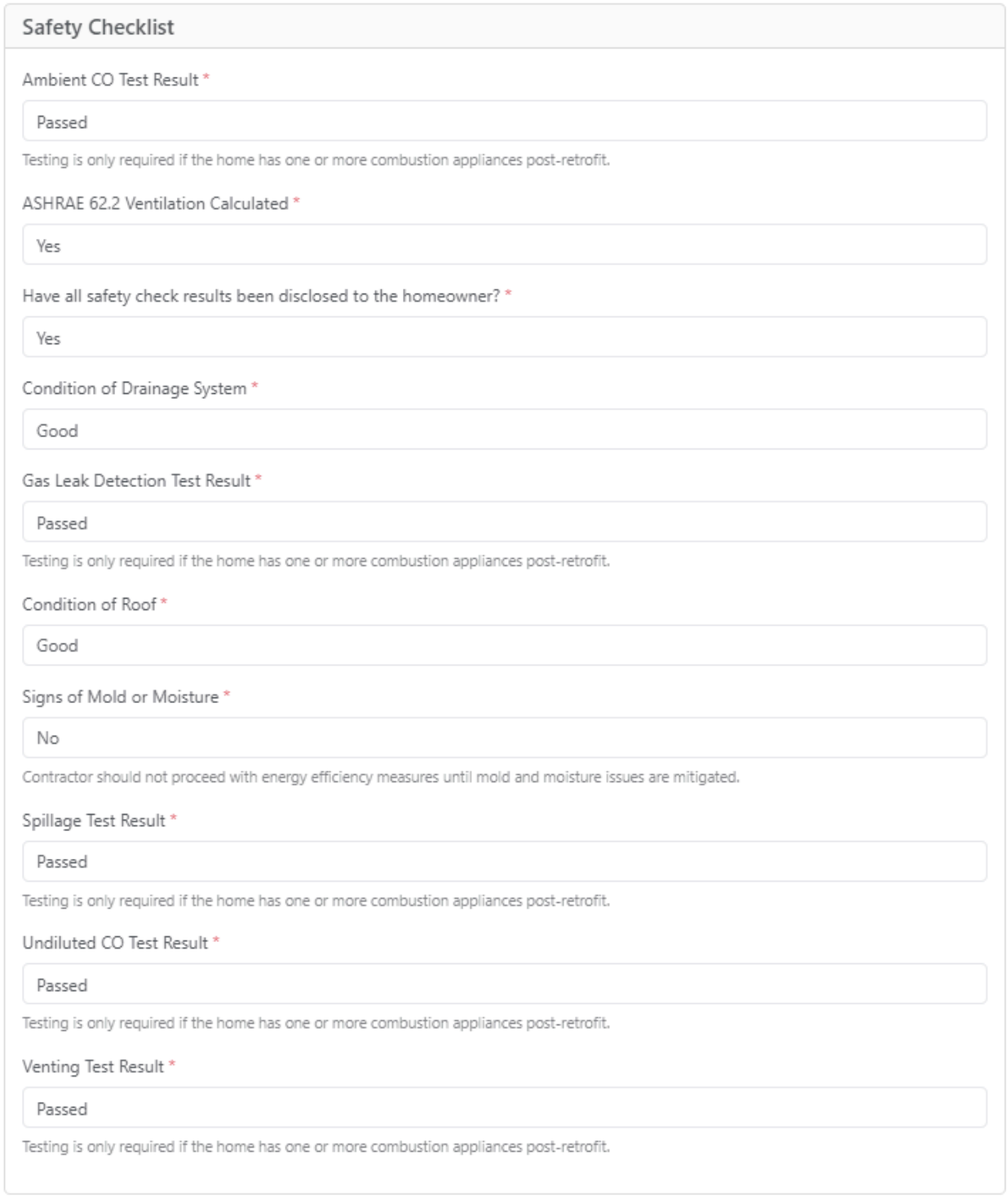
Line Items

- Materials (Min 1 Req’d)

    - Name
    - ID/SKU
    - Cost
- Equipment (Min 1 Req'd)
    - Name
    - ID/Model
    - Cost
    - UPC (For heat pump measures: only model numbers(s) or AHRI number will be accepted)
    - Count
    - AHRI Reference number
    - Description
    - Purchase Date
    - Labor Cost

Safety Checklist (fill out here not in QIT)

Fill this section out for all rebates, HOMES and HEAR.



Safety Checklist

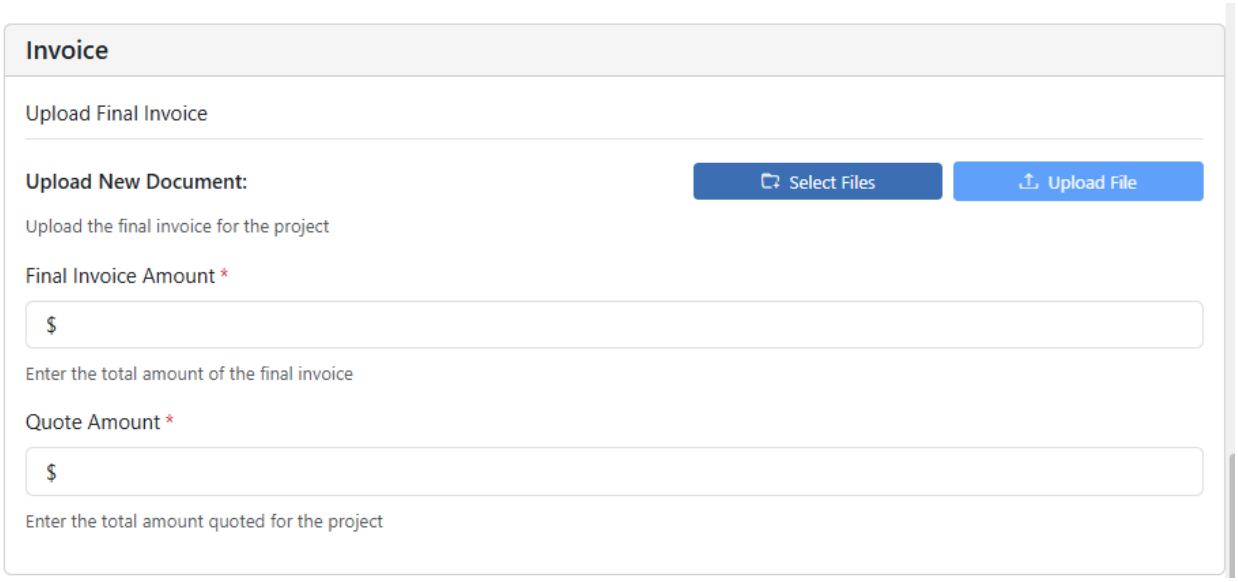
- Ambient CO Test Result:
    - Passed
    - Failed
    - Not Applicable

    - Warning
- ASHRAE 62.2 Ventilation Calculated:
    - Yes
    - No
    - Not Applicable
- Have all safety check results been disclosed to the homeowner?
    - Yes
    - No
    - Not Applicable
- Condition of Drainage System:
    - Good
    - Potential Issues
    - Not Applicable
- Gas Leak Detection Test Result:
    - Passed
    - Failed
    - Not Applicable
    - Warning
- Condition of Roof:
    - Good
    - Potential Issues
    - Not Applicable
- Signs of Mold or Moisture:
    - Yes
    - No
    - Not Applicable
- Spillage Test Result:
    - Passed
    - Failed
    - Warning
    - Not Applicable
- Undiluted CO Test Result:
    - Passed
    - Failed
    - Warning
    - Not Applicable
- Venting Test Result:
    - Passed
    - Failed

- Warning
- Not Applicable

Invoice

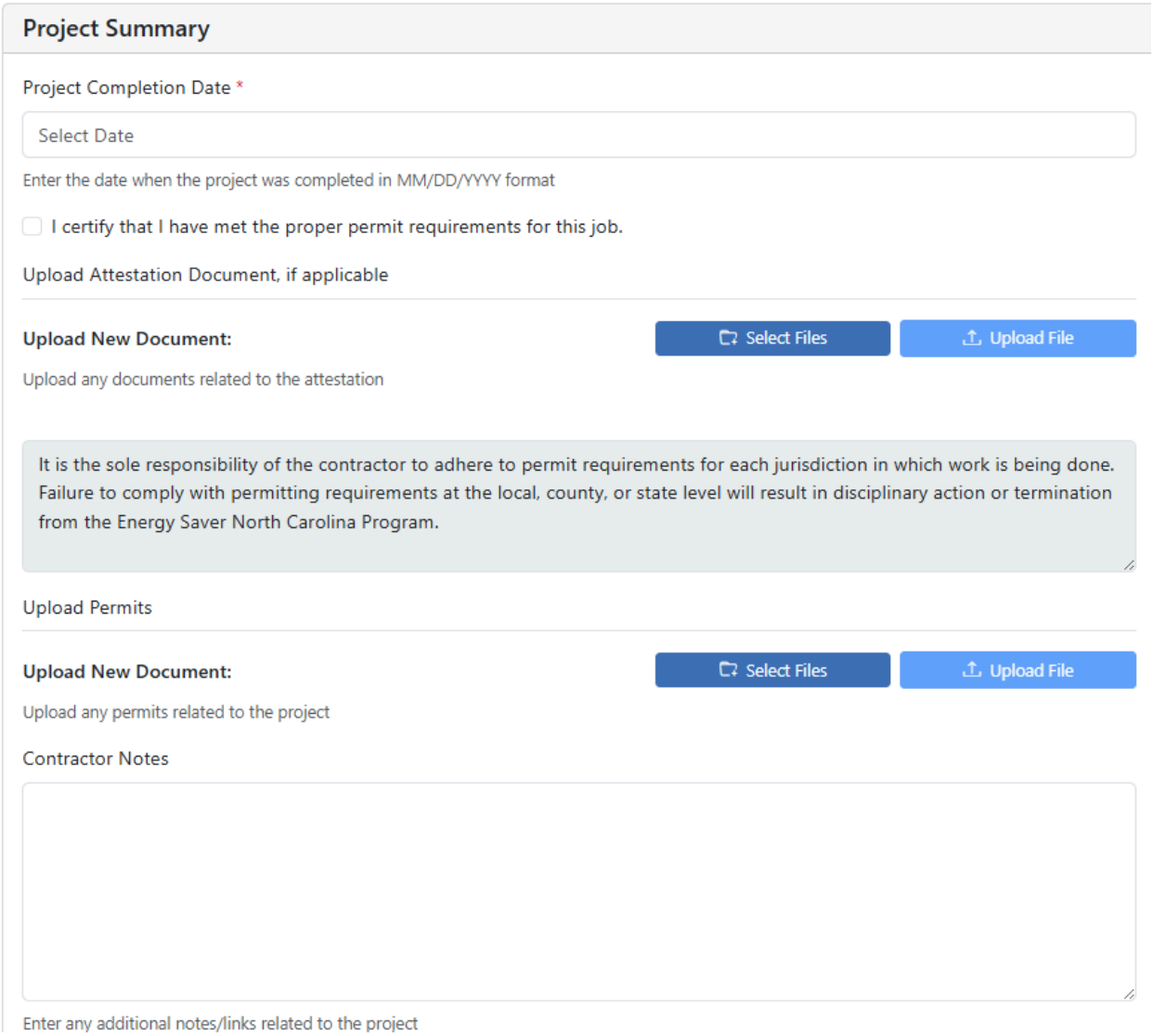
Fill this section out for all rebates, HOMES and HEAR. Please lump fees and taxes into the equipment cost field so that totals from invoice match.



- Upload Final Invoice Allows you to continue without upload, cannot change once submitted, if skipped this will result in delayed rebate processing
- Final Invoice Amount
- Quote Amount

Project Summary

Fill this section out for all rebates, HOMES and HEAR.



- Project Completion Date
- Check box: I certify that I have met the proper permit requirements for this job.
- Upload Attestation Document
- Upload Permits, if applicable
- Contractor notes

Hit Next Button

Appendix A: Portal Status Meanings

Status:	Meaning:
Pre-qualification Pending Submission – User Action Required	Customer application is incomplete
Pre-qualification Review – In Progress	Customer application has enough information for program staff to begin review
Pre-qualification Returned – User Action Required	Customer application returned to customer, customer needs to submit additional documentation
Assessment – Pending Scheduling (HOMES Only)	Customer application approved, pending scheduling of assessment
Assessment – Scheduled (HOMES Only)	Assessment has been scheduled
Project Application – Pending Submission	The customer application is open to the contractor and awaiting their submission of the rebate reservation
Project Application – Review	Contractor's application for Rebate Reservation is being reviewed by program staff
Project Application Returned	The application for rebate reservation was returned to the contractor for more information
Project Approved – Rebate Reserved	The application for rebate reservation was approved and the rebate was reserved
Post Project Review – In Progress	The final Rebate Application from the contactor is being reviewed by program staff
Post Project Review – Returned	The final Rebate Application from the contactor has been returned to the Contractor for additional information
Project Completed	The Rebate Application is approved and ready to send to payment
Payment Initiated	Payment has been requested from Choice Digital
Rebate Paid	Payment has been issued
Transferred	**Not used** If you see something in this status please contact program staff
Wait Listed	The customer is waitlisted due to phase/program not open
Cancelled	Applicant does not qualify or has opted out of program

Appendix B: Invoice Guidance

- Descriptions of work should be descriptive enough that it's clear what work was performed/authorized for the price (See below).
- If an Approved Contractor reduces their normal price to be equal to the amount of the rebate to avoid out-of-pocket costs to the customer, then the bid/invoice should show the normal amount for that line-item and the discount applied to that line-item.
- Ensure the rebate does not exceed the total project cost when combined with other permissible/eligible program rebates accepted by the eligible rebate recipient (i.e., total rebate funding including state, federal, local, and utility rebates must not exceed the total project cost).
- Ensure that rebates are not applied to specific energy efficiency upgrades funded with other Federal grants or rebates (e.g. can't split cost of an insulation upgrade with a federally funded program like the Weatherization Assistance Program and the IRA program).
- Ensure that upgrades receiving rebates benefit only the residential portion of a mixed-use building. Upgrades that benefit any portion of the commercial space are not eligible for rebate.
- If a project scope changes (for example, the applied for product is out of stock) you must ensure that any replacement product still meets the program requirements. **Unqualified changes may not be eligible for rebate dollars.**
- No Mandatory Arbitration Clauses: Contracts associated with the IRA Section 50121 & 50122 program cannot contain clauses that mandate arbitration, ensuring that consumers have the right to seek judicial review if needed.
- Contracts will incorporate the holder-in-due-course rule to ensure consumer protections remain intact, even if the contract is assigned to a third-party creditor.

HOMES and HEAR bids need to be separate and all bids need the following:

- Customer name
- Installation Address
- Contractor name
- Contractor contract information (address, phone, and email)
- Bid date (for reservation invoice) OR Date of Completion (for closeout invoice)
- Program : HOMES or HEAR
- Total Price (for final invoice at closeout we require equipment, materials and labor costs to be broken out into separate line items.)
- Discounts

- Other incentives (utility) You do not need to include potential mail-in rebates on your invoice.
- HOMES or HEAR Rebate amount (per measure for HEAR)
- Customer out of pocket total
- Signed and dated by the Property owner or their authorized representative (such as a property manager acting on behalf of an owner).

Measure Specific Requirements:

Heat Pump Water Heater

- AHRI #
- Brand and description
- Model #
- Efficiency
- Tank size
- Description of installation labor
- Price

Heat Pump

- AHRI #
- Outdoor unit
    - Brand and description
    - Model #
    - HSPF2
    - SEER2
    - Heating capacity
    - Cooling capacity
    - Description of install labor
- Compatible indoor unit description for each outdoor unit
    - Brand and description
    - Model #
    - Description of install labor
- Price

Electric Heat Pump Dryer

- Model number
- Unit description
    - Brand
    - type of appliance
- Description of labor involved

- Price

Electric Stove, Cooktop, Range, or Oven

- Model number
- Unit description
    - Brand
    - type of appliance
- Description of labor involved
- Price

Electric Load Service

- Detailed description
    - Materials needed
    - Is upgrading service needed?
    - Which qualified equipment install requires panel upgrade
- Description of labor involved
- Price

Electric Wiring

- Detailed description
- Materials needed
- Which qualified upgrade install requires electrical wiring
- Description of labor involved
- Price

Insulation

- Description of insulation
    - Location of insulation to be installed.
    - Square feet of area to be insulated
    - Type of Energy Star insulation
    - R-value added
    - Ending R value
- Description of associated insulation upgrades. Examples include but not limited to (reference NC Weatherization Installation Standard Work Specifications)
    - Air sealing prior to insulation
    - Add adequate passive ventilation
    - Terminate exhaust fans to dedicated ports to outdoors
    - Appropriate baffles for ventilation and insulation barriers

    - Seal and insulate hatch
    - Vapor barriers covering ground in crawlspace
- Description of labor involved
- Price

Ventilation

- Unit description (brand, type of ventilation)
- Energy Star Unique ID
- Description of labor, including confirmation it will be exhausted to the outdoors.
- Price
- *Ventilation materials are eligible for rebates only in homes that also receive air sealing services or otherwise achieve low air leakage rates (e.g., 3 ACH50 per International Energy Conservation Code (IECC)).

Air sealing

- List of each air sealing location (attic, crawlspace, basement, etc.).
- Description of which penetrations and air infiltration will be addressed
- Description of product
- Description of labor involved.
- Price

Duct sealing

- List of each duct sealing location (attic, crawlspace, etc.).
- Description of product
- Description of labor involved.
- Price

Windows

- Brand and type of windows
- U Value and SHGC
- Number of windows
- Sq ft of each window
- Side of the house (N, W, S, E)
- Description of labor involved.
- Price